SEQUENTIAL FUNCTION CHART Workshop User's Guide

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Sequential Function Chart Workshop User's Guide for Automation Studio™.

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1 Introduction

Automation Studio™ SFC is a complete automation project specification environment. It is a direct manipulation graphic editing tool as well as a SFC simulation tool. The notation used conforms to the CEI 61131-3 International Standard and to the extension proposals as defined by the AFCET's SFC group and by the PR NF EN 60848 standard projects. Because Sequential Function Chart (SFC) is rapidly becoming the method of choice to design structured automation controls, the Automation Studio™ SFC Workshop provides all of the necessary tools to design operation sequences and to validate a project prior to implementation, as well as to understand how equipment operates.

This *SFC Workshop User's Guide* provides the necessary information to use this workshop with Automation Studio™. This includes technical specifications for components, procedures for defining properties, building and simulation of a circuit, and examples of applications.

This guide is divided into the following sections:

Chapter *2 SFC Representation*: This chapter describes the representation of an element constituting a typical SFC and its illustration in Automation Studio™.

Chapter *3 Description of the SFC Editor*: This chapter presents the SFC Workshop environment and describes the editable properties of various SFC components.

Chapter *4 SFC Syntax*: This chapter gives a detailed description of the syntax that must be applied while using SFC components, symbols and processes.

Chapter *5 Predefined SFC Variables*: This chapter describes the function and utility of variables that are automatically created with the creation of a SFC and its components.

Chapter *6 Creation of a First SFC:* In this chapter an example of an application is presented as a small project, in order to assimilate all the functionalities of this workshop.

Chapter *7 Simulation*: This chapter describes the particularities associated with the simulation of SFCs. Its purpose is to familiarize the user with the simulation mode of the workshop.

2 SFC Representation

An automation system is generally comprised of two independent parts:

- The Operative Part (OP): the system itself, that the user wishes to automate;
- The Command Part (CP) : the logical unit that sends OUTPUT signals to the operative part or other external components, given a set of INPUT signals.

A Sequential Function Chart (SFC) is a graphical formalism that allows the description of an automation system, using basic SFC components,

The basic elements are:

- Steps, with which actions are associated;
- Transitions, with which conditions are associated;
- Structured texts, or actions that are not linked to a step;
- Links to connect steps and transitions.

2.1 Graphic Representation

The following figure represents a typical SFC using various basic SFC components.

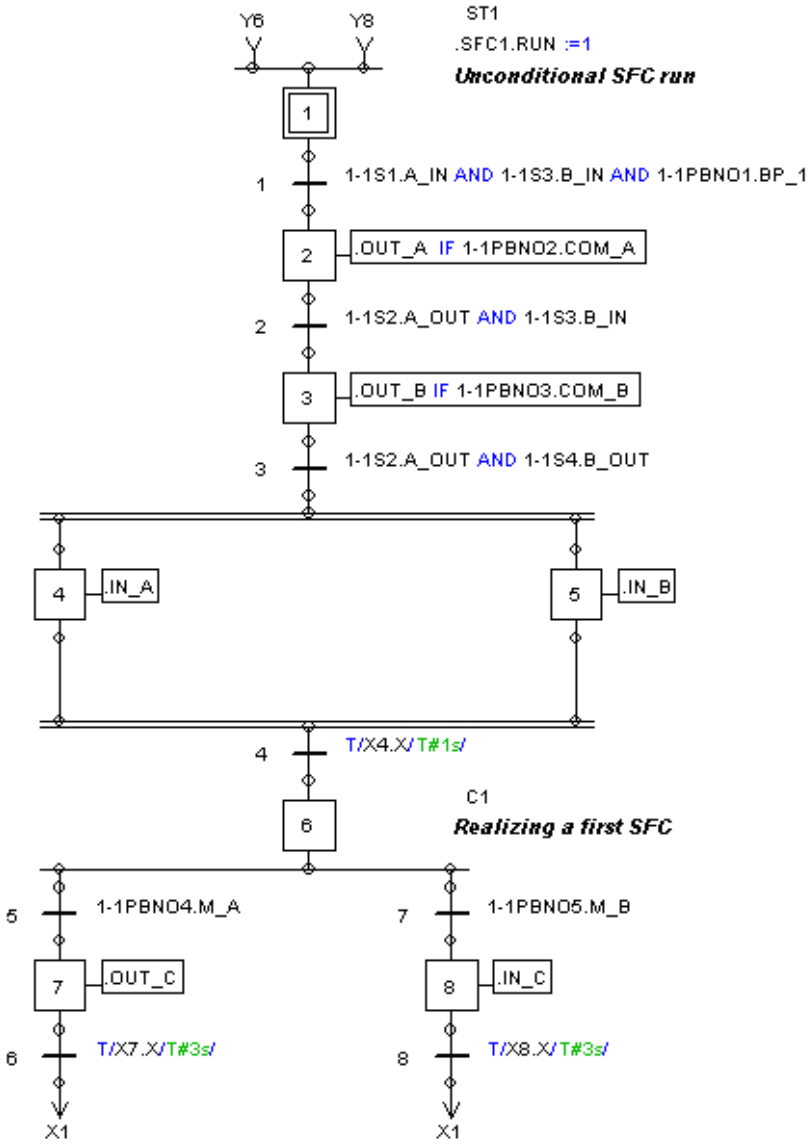
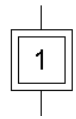
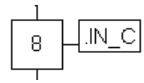
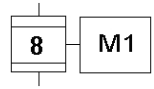
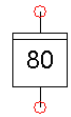
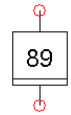
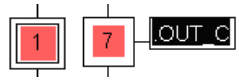
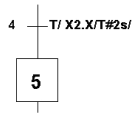
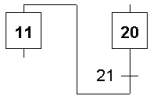
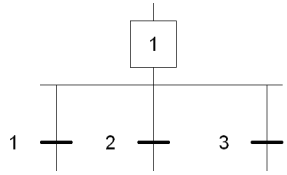
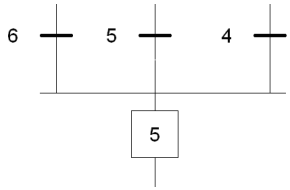
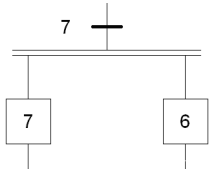


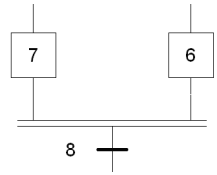
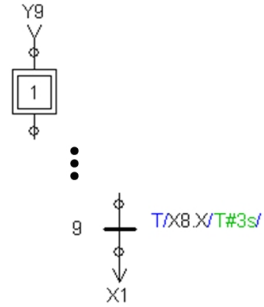
Figure 2-1: SFC Example

2.2 Description of SFC Components

The elements that can be found in a SFC are listed in the following table:

Name	Symbol	Description
Inactive Initial Step		A double box identified by a number represents an initial step.
Standard Step an Action		The input step is always on top. The output step is always on the bottom. A box identified by a number represents a standard step. Actions are connected to their corresponding step with a link.
Macro Step		A box with rectangles at the top and bottom and identified by a number represents a macro step. The name of the SFC that is activated by the macro step is in a box on the right side of the macro step.
In Step		A box with a rectangle on top and identified by a number represents an IN step.
Out Step		A box with a rectangle on the bottom and identified by a number represents an OUT step.
Active Steps		An active step is represented by a red token placed in the centre of the step symbol during simulation.

Name	Symbol	Description
Transition		A transition is represented by a cross joining two steps. A numerical marker identifies a transition. The transition condition is always positioned on the right side of the transition symbol.
Source Transition		A source transition is a transition that is not linked to an upstream step.
Link		A link connects the output of transition 21 to the input of step 11.
OR Divergence		When a connection exists between one step and more than one transition, an OR divergence is drawn instead of the link.
OR Convergence		When a connection exists between multiple source transitions and one step, an OR convergence is drawn instead of the link.
AND Divergence		When a connection exists between a single source transition and multiple steps, an AND divergence is drawn instead of the link.

Name	Symbol	Description
AND Convergence		When a connection exists between multiple source steps and one transition, an AND convergence is drawn instead of the link.
Structured Text	**ST3** E := C AND D	Structured texts are actions that are independent and not linked to a step; they are linked to a SFC.
Comment	**C25** ***When the operator pushes the START button, the initial conditions must be present.***	Comments can be positioned anywhere in the diagram, they are bound to the SFC.
Jumps		Jumps simplify SFC diagrams. Jumps replace links between two components that are not close to each other.
		Indicates that the link goes to component 1.
		Indicates that the link comes from component 9.

3 Description of the SFC Editor

This chapter covers the SFC editor. All descriptions of other items with common functions can be found in the *User's Guide*.

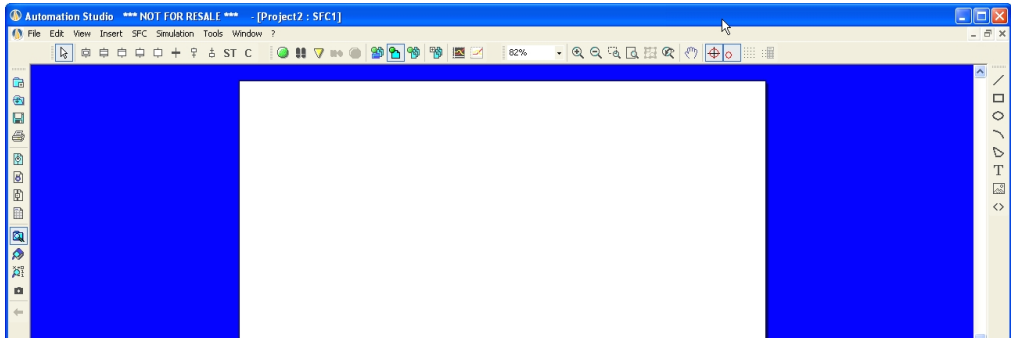


Figure 3-1: SFC Editor's Main Window

3.1 "Edit" Menu

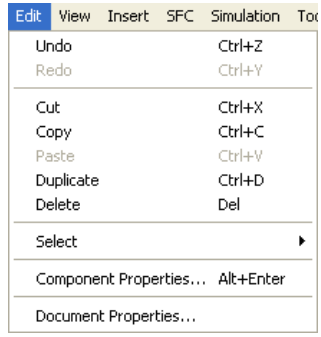


Figure 3-2: "Edit" Menu

The SFC Editor's Edit menu options are as follows:

Select

Allows the selection of every step, transition, structured text, comment, or link; everything that follows a selected transition/step, and every item of a SFC.

The options available from the Select command in the Edit menu are:

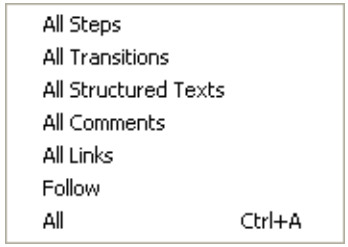


Figure 3-3: Options from the Select Command in the "Edit" Menu

Option	Description
All Steps	Selects all steps.
All Transitions	Selects all transitions.
All Structured Texts	Selects all structured texts.
All Comments	Selects all comments.
All Links	Selects all links.
Follow	Selects everything that follows the currently selected step or transition (including itself).
All	Selects all components in the SFC.

Component Properties

This allows the accessing of the properties of the selected component.

Document Properties

This allows the accessing of the document properties.

Branch	Description
Page Setup	Same as other document types

Branch	Description
Map Locator	Same as other document types
Summary Info	Same as other document types
SFC	Configuration of the SFC
History	Same as other document types

3.2 Component Properties

Most of the branches in the "Document Properties" dialog box are similar to the same branches in other document types. They are described in the *Automation Studio™ User's Guide*. The "SFC" branch is described in the next section.

3.2.1 "SFC" branch

This dialog box allows you to set the parameters of the SFC environment

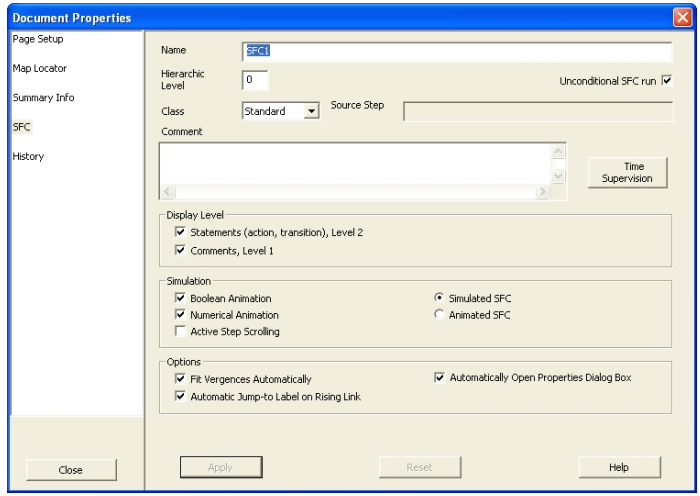


Figure 3-4: "Document Properties" Dialog Box; "SFC" Branch

Section	Description
Name	The default SFC name is displayed in this zone. This field can be modified. It is synchronized with the name of the document that appears in the Project Explorer.
Hierarchic Level	This zone allows the modification of the SFC hierarchic level. The level is 0 by default (highest level). The hierarchic level must be in the 0 to 999 range.
Class	This allows you to specify whether the SFC is an explosion of a macro step or a standard SFC by using the drop-down list . The SFC is defined as being STANDARD by default.
Source Step	The source macro step of the macro step SFC is defined in this zone.
Unconditional SFC Run	This checkbox, checked by default, allows an unconditional running of the SFC. If the checkbox is unchecked, a structured text must contain the SFC execution condition. For example: *SFC_Name*.RUN IF CONDITION
Comment	You may write a general comment for the SFC in this zone.
Time Supervision	This button gives you access to the “Sequence Time Supervision dialog box”. The sequence time is calculated from the start time to the return to the initial step. To use this function, the SFC must have step 1 as the only initial step.
Display Level	This zone allows you to specify what information will be displayed for the SFC components.
Comments, Level 1	Displays the SFC components’ comments (Level 1).
Statements (action, condition), Level 2	Displays the SFC components’ statements (action and condition) (Level 2).

Section	Description
Simulation	This zone allows you to specify the behaviour of the SFC while in the simulation mode.
Boolean Animation	When this option is selected, the names of the Boolean variables with TRUE conditions are displayed in reverse video during a simulation.
Numerical Animation	When this option is selected, during a simulation, a field containing the variable's value is displayed on the right of the numerical variable's name.
Active Step Scrolling	When this option is selected, the SFCs are moved inside their own window in order for the active steps to remain visible on the screen. The display priority is granted to the SFCs' upper left side steps. So, when two steps of a single SFC are simultaneously active, the top priority is granted to the top left-most step of the SFC.
Simulated Flow Chart	When this option is selected, the simulator executes the actions and evaluates their links. In other words, it updates the variables and it simulates the flow chart.
Animated Flow Chart	When this option is selected, the simulator simply reads and describes the state of each internal and external variable. (OPC) In other words it displays the state of the variables and it simulates the flow chart.
Options	This zone allows to set some default properties of the SFC environment.
Fit Vergences Automatically	This option adjusts AND and OR links based on the shortest path between the step/transition components.
Automatic Jump-to Label on Rising Link	This option automatically replaces rising links with jumps in order to simplify SFCs.

Section	Description
Automatic Open Properties Dialog Box	When chosen, this option automatically opens the properties dialog box of the component (SFC, step, transition, comment, or structured text).

3.2.2 Sequence Time Supervision

In this dialog you can set limits on the SFC's sequence time.

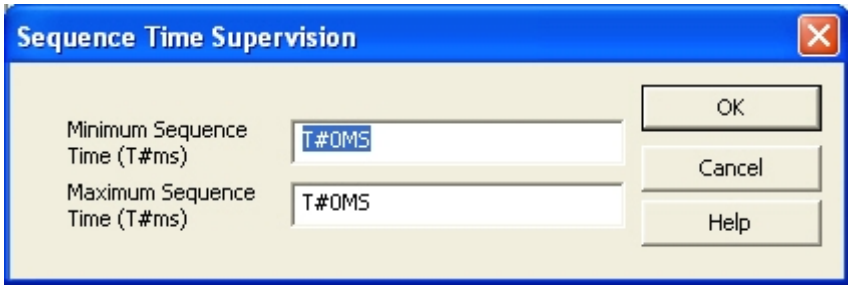


Figure 3-5: "Sequence Time Supervision" Dialog Box

Fields	Description
Minimum Sequence Time (T#0MS)	This option allows you to set a minimum time for the SFC's sequence. If the sequence is active for a sorter period, the minimum sequence time defaults to TRUE (eg. : SFC1.SEQ_MIN_FAULT).
Maximum Sequence Time (T#0MS)	This option allows you to set a maximum time for the SFC's sequence. If the sequence is active for a longer period, the maximum sequence time defaults to TRUE (eg. : SFC1.SEQ_MAX_FAULT).

3.3 "SFC" Menu

The commands that are available in the "SFC" menu are as follows:

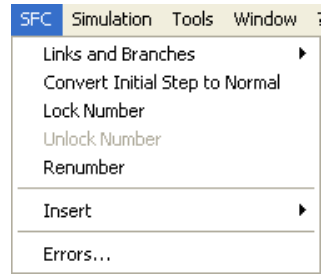


Figure 3-6: "SFC" Menu

Links and Branches

This provides access to actions on links and branches. (AND or OR convergence/divergence).



Figure 3-7: Menu "SFC", command "Links and Branches"

The first two options can also be accessed through the contextual menu of a vergence. The last option is also available through the contextual menu of a link or jump.

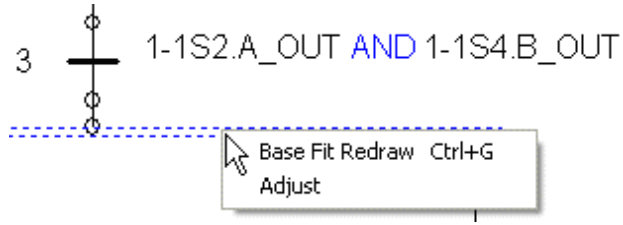


Figure 3-8: Contextual Menu for a vergence

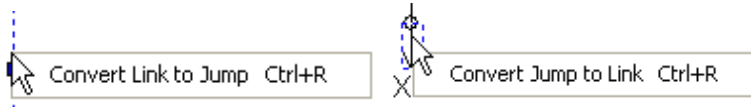


Figure 3-9: Contextual Menus for a Link and a Jump

The "Links and Branches" command in the "SFC" Menu offers the following options:

Option	Description
Base Fit Redraw	Redraws the selected vergence and its connected links using the shortest path between the components.
Adjust	Redraws the selected vergence with a jump using the shortest path.
Convert Link	Converts a Link into a Jump, or vice versa according to the context..

Convert Step

This converts a standard step into an initial step and vice versa.

Lock Number

This protects the selected item number from being renumbered.

Unlock Number

This opens the selected item number to be renumbered.

Renumber

This renumbers all selected unlocked items.

Insert

This grants access to the component insertion tool in the SFC. Components can be inserted from the toolbar.

The available options from the Insert command in the SFC menu are as follows:

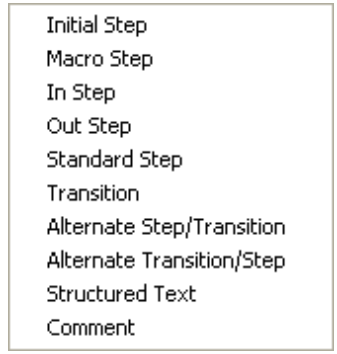


Figure 3–10: Options from the Insert Command in the "SFC" Menu

Option	Description
Initial Step	Insert an initial step.
Macro Step	Insert a macro step.
In Step	Insert an initial step on a SFC defined a macro.
Out Step	Insert an exit step on a SFC defined as a macro.
Standard Step	Insert a normal step.
Transition	Insert a transition.
Alternate Step/Transition	Alternate the insertion of normal steps and transitions. Start with a step.
Alternate Transition/Step	Alternate the insertion of transitions and normal steps. Start with a transition.
Structured Text	Insert structured text.
Comment	Insert a free comment.

Errors

This allows you to identify errors in the SFC. Error and/or warning messages are displayed in the Messages Window.

3.4 The "SFC" Toolbar

The "SFC" toolbar contains all the buttons that correspond to the most utilized commands from the Project Explorer and Diagram Editor. The toolbar automatically adjusts itself to the open utility. When the command is unavailable in the utility, the button is greyed out and therefore inaccessible.



Figure 3-11: "SFC" Toolbar

The tools available in the "SFC" toolbar are as follows:

Button	Tool	Description
	Selection	Returns to the original arrow pointer.
	Initial Step	Inserts an initial step.
	Macro step	Inserts a macro step.
	In Step	Inserts an IN step in a macro defined SFC
	Out Step	Inserts an OUT step in a macro defined SFC.
	Standard Step	Inserts a standard step in a macro defined SFC.
	Transition	Inserts a transition.

Button	Tool	Description
	Alternate Step/Transition	Inserts an alternate standard step/transition.
	Alternate Transition/Step	Inserts an alternate transition/standard step.
ST	Structured Text	Inserts a structured text.
C	Comment	Inserts a free comment.

3.5 Step Properties

In this dialog the user can specify the equations that will determine the actions that are associated with a step.

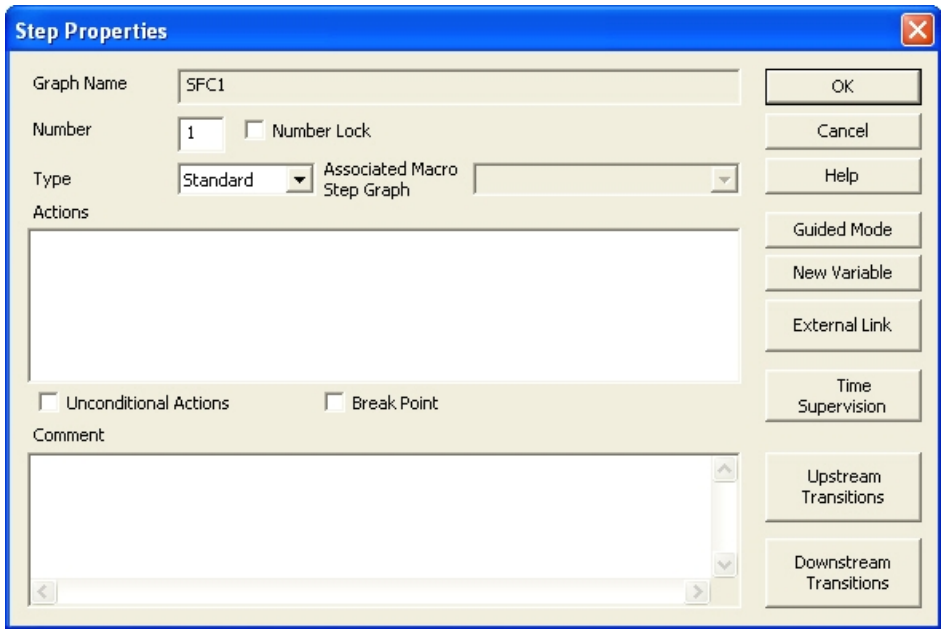


Figure 3-12: "Step Properties" Dialog Box

Zones	Description
Graph Name	In this zone, the name of the SFC containing the step appears. In this dialog box, this field is non-modifiable. However, it can be modified in the "Name" field of the "SFC" branch in the "Document Properties" dialog box.
Number	A number is automatically attributed to each new step. This number can be modified as long as it is within the 1 to 999 range. Two steps in the same SFC cannot have the same number.
Number Lock	If this option is selected, the number of the step will not be modified during an automatic renumbering of the SFC. When the number has been locked, an asterisk (*) appears at the center of the left side of the exterior frame.
Type	Type of step. The user can choose between standard, initial, macro, in or out type from a drop-down list.
Associated Macro Step Graph	If the step is defined as being a macro step, the associated SFC that will activate the macro step must be defined in this zone. This field is deactivated if the step is not defined as a macro step.
Actions	This zone is used to edit the different actions associated with the step. Here, all of the actions associated with a step can be visualized. This zone is deactivated for macro steps and out steps. The user may: ● Enter multiple actions by separating them with a semicolon; ● Enter multiple actions submitted to the same condition by separating them with a comma. (ex.: Action1, Action2, Action3 IF Condition1). The write action to be executed by using the variables and operators can be found in the Guided Mode dialog box.

Zones	Description
Guided Mode	This button provides access to the Guided Mode dialog box. This button is greyed out for macro steps and out steps.
New Variable	This button provides access to the Add a Variable dialog box. For more details on creating variables, see the variables section in the *Automation Studio™ User's Manual*.
External link	This button provides access to the OPC Settings dialog box to create external links (OPC, etc.). For more details on creating external links, see the variables section in the *Automation Studio™ User's Manual*.
Unconditional Action	This checkbox determines whether the running of the actions of the step is conditional to the running state of the SFC.
Break Point	When checked, this option will stop the simulation when the step has been activated.
Time Supervision	This button provides access to the Step Time Supervision dialog box. The sequence time is calculated from the start to the return to the initial step.
Upstream Transitions	This button permits the visualization of the complete list of transitions preceding the step.
Downstream Transitions	This button permits the visualization of the complete list of transitions succeeding the step.
Comment	This button allows you to add comments related to the action.

3.5.1 Step Time Supervision

This dialog allows the user to specify minimum and maximum activation times for the step.

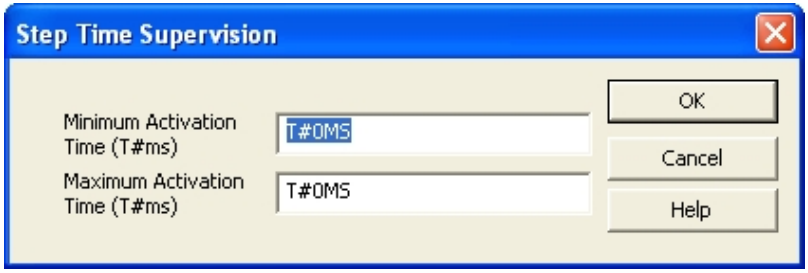


Figure 3-13: "Step Time Supervision" Dialog Box

Zone	Description
Minimum Activation Time (T#0ms)	Sets a minimum time for the step's activation. If the step is active for a shorter period than the one specified here, the minimum activation time for the step defaults to TRUE. (eg.: SFC1.STEP_MIN_FAULT) If the entered value is greater than zero, it must be smaller than the maximum activation time.
Maximum Activation Time (T#0ms)	Sets a maximum time for the step's activation. If the step is active for a period longer than the one specified, the maximum activation time for the step defaults to TRUE (eg.: SFC1.STEP_MAX_FAULT). If the entered value is greater than zero, it must be greater than the minimum activation time.

3.5.2 Upstream and Downstream Transitions

These dialog boxes show the upstream or downstream transitions from the step.

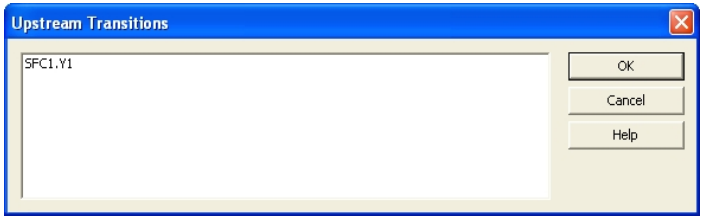


Figure 3-14: "Upstream Transitions» Dialog Box

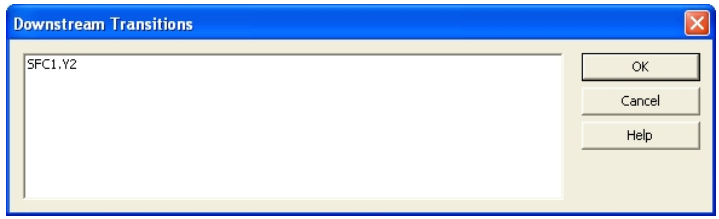


Figure 3-15: "Downstream Transitions» Dialog Box

3.6 Guided Mode for the Step

This dialog box contains an interface that allows you to build the equations for a step, as an alternative to you directly typing in those equations. Every possible action or equation corresponds to a button in the interface. The dialog box displays a basic form of the equation that is updated whenever the user selects another action. Some of the fields of the equation should be replaced by more appropriate (numerical) values.

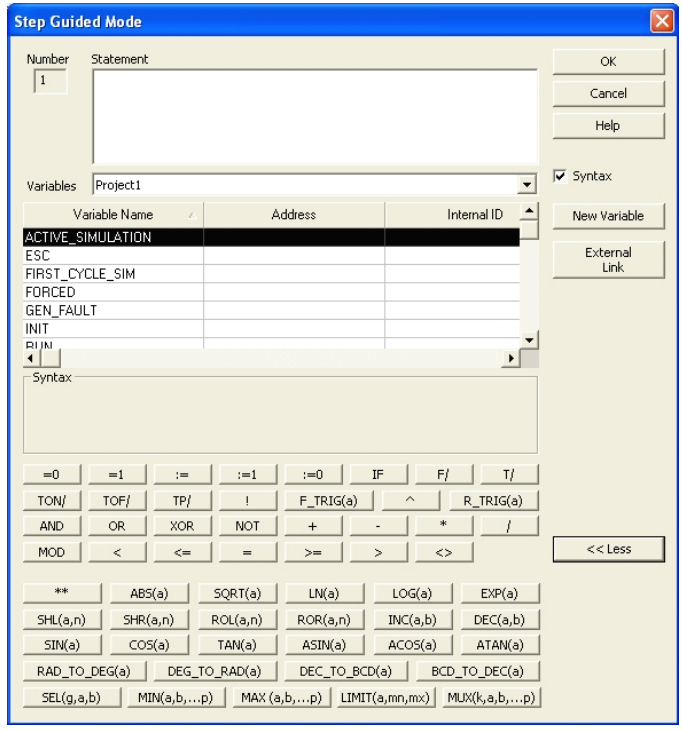


Figure 3-16: "Step Guided Mode" Dialog Box

Zone	Description
Number	Displays the step number during edition.
Statement	This zone is used to type or edit the actions associated with the step. The syntax of the actions must follow that described in chapter *4 SFC Syntax*. Below is an example of a typical action. **Basic Equation Form** TON/Xn.X/T#a/ **Edited Equation** TON/X1.X/T#23s/
Variables	This drop-down list allows the selection of variables that are global (the entire project) or local (current diagram).
Variable Name	This table contains the variable names with the selected type of association. Double-clicking on a variable selects it and adds it to the action statement. The left side column displays the variable name, and the right side column, displays the associated component, if this is the case. See the *Variables Manager* in the *Automation Studio™ User's Guide*.
Syntax	Describes the operator that has been selected by a left clicking in the examples list, or in the calculator.
Syntax (check box)	This check box will activate syntax analysis during edition.
New variable	Accesses the Add a variable dialog box. For more details on creating variables, see the *Variables Manager* in the *Automation Studio™ User's Guide*.
External link	Accesses the "OPC Settings" dialog box to create external links (OPC, etc.). For more details on creating external links, see the *Variables Manager* in the *Automation Studio™ User's Guide*.

Zone	Description
More/Less	Use this button to gain access to the advanced features of the operator calculator.
Operators	These buttons are used to enter operators in the action statement. Click on the corresponding button to enter an operator.

3.7 Transition Properties

The user can specify the conditions associated with a transition in this dialog box.

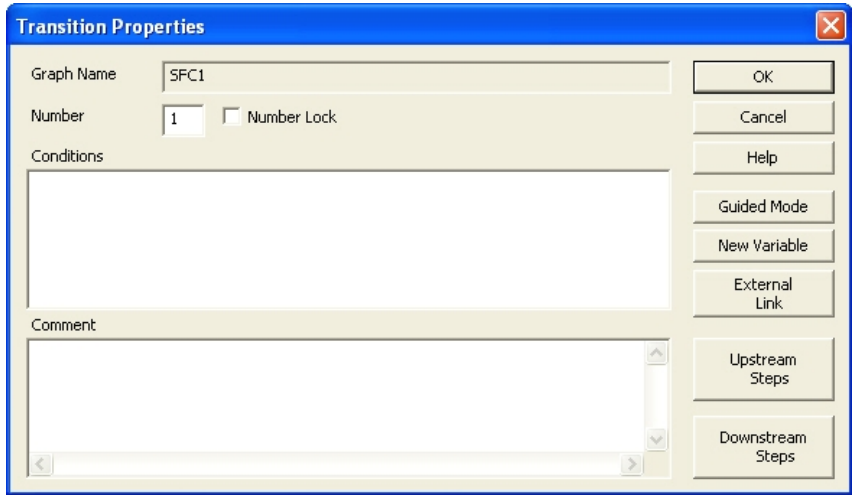


Figure 3-17: "Transition Properties" Dialog Box

Fields	Description
Graph Name	In this zone, the name of the SFC containing the transition appears. In this dialog box, this field is non-modifiable. However, it can be modified in the Name field of the SFC branch in the "Document Properties" dialog box.

Fields	Description
Number	A number is automatically attributed to each new transition. This number can be modified as long as it is within the 1 to 999 range. Two transitions in the same SFC cannot have the same number.
Number Lock	If this option is selected, the number of the transition will not be modified during an automatic renumbering of the SFC or sections of the SFC. When the number has been locked, an asterisk (*) appears to the right of the transition number.
Conditions	This zone is used to edit the transition condition.
Comment	Comments may be added in reference to the transition.
Guided Mode	This button opens the "Guided Mode" Window to edit the transition condition. The transition condition must conform to the SFC syntax as per Chapter *4 SFC Syntax*.
New Variable	Accesses the "Add a variable" dialog box. For more details on creating variables, see the *Variables Manager* in the *Automation Studio™ User's Guide*.
External Links	Accesses the OPC Settings dialog box to create external links. (OPC, etc.) For more details on creating external links, see the *Variables Manager* in the *Automation Studio™ User's Guide*.
Upstream Steps	This button permits the visualization of the complete list of steps preceding the transition.
Downstream Steps	This button permits the visualization of the complete list of steps succeeding the transition.

A transition with no transition condition is considered false. During simulation, only the contextual menu permits passing a transition with no transition condition.

3.8 Structured Text Properties

In this dialog box, the user can specify the equations that determine the actions associated with a structured text.

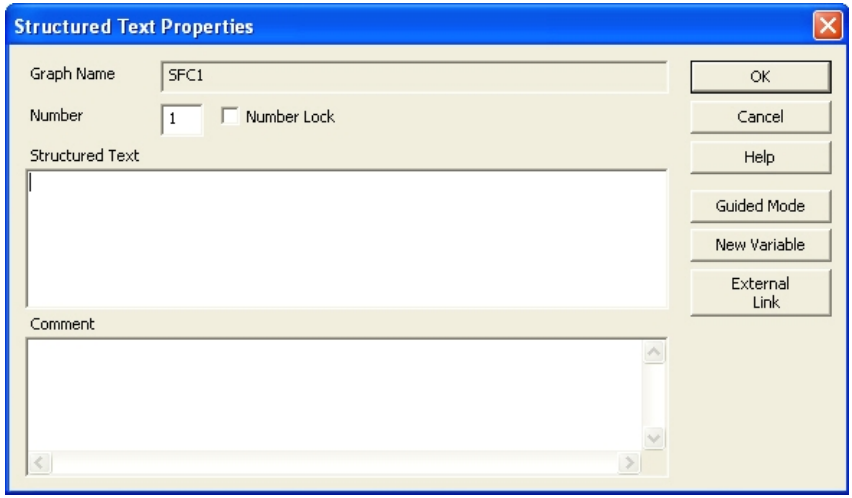


Figure 3-18: "Structured Text Properties" Dialog Box

The zones and buttons of this dialog box are:

Zones	Description
Graph Name	The name of the SFC that contains the structured text. In this dialog box, this field is non-modifiable. However, it can be modified in the Name field of the SFC branch in the "Document Properties" dialog box.
Number	A number is automatically attributed to each new structured text. This number can be modified as long as it is within the 1 to 999 range. Two structured texts in the same SFC cannot have the same number.
Number Lock	If this option is selected, the number of the structured text will not be modified during an automatic renumbering of the SFC or of sections of the SFC. When the number has been locked, an asterisk (*) appears to the right of the transition number.

Zones	Description
Structured Text	Use this zone to define the structured text statement. Structured texts must conform to SFC syntax guidelines as defined in Chapter *4 SFC Syntax*.
Comment	Add comments in reference to the structured text.
Guided Mode	This button accesses the Structured Text Guided Mode dialog box to edit the structured text.
New variable	This button accesses the Add a variable dialog box. For more details on creating variables, see the *Variables Manager* in the *Automation Studio™ User's Guide*.
External Link	This button accesses the OPC Settings dialog box to create external links (OPC, etc.). For more details on creating external links, see the *Variables Manager* in the *Automation Studio™ User's Guide*.

3.9 Comment Properties

In this dialog box the user can edit the text of a comment.

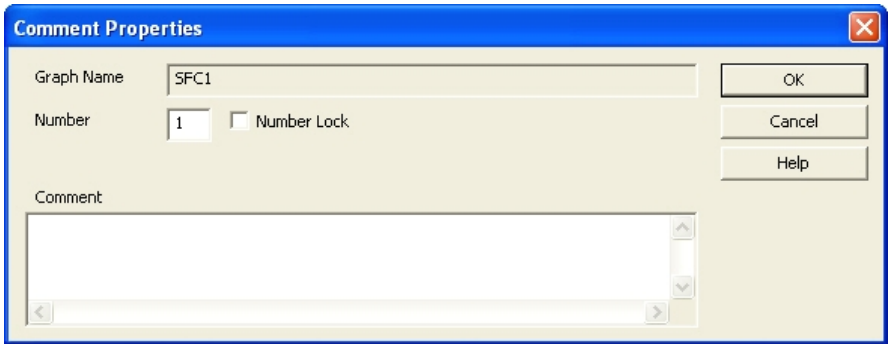


Figure 3-19: "Comment Properties" Dialog Box

The zones and buttons in this dialog box are:

Zones	Description
Graph Name	This is the name of the SFC that contains the comment. In this dialog box, this field is non-modifiable. However, it can be modified in the "Name" field of the SFC branch in the "Document Properties" dialog box.
Number	A number is automatically attributed to each new comment. This number can be modified as long as it is within the 1 to 999 range. Two comments in the same SFC cannot have the same number.
Number Lock	If this option is selected, the number of the comment will not be modified during an automatic renumbering of the SFC or sections of the SFC. When the number has been locked, an asterisk (*) appears to the right of the transition number.
Comment	Add comments.

3.10 SFC Environment

The "Options" Menu has two branches that allow you to modify the SFC environment. The first branch concerns the behaviour while in the edit mode, while the second branch concerns the behaviour while in the simulation mode.

3.10.1 SFC Editor

You can modify the default behaviour of the SFC editor through this dialog box.

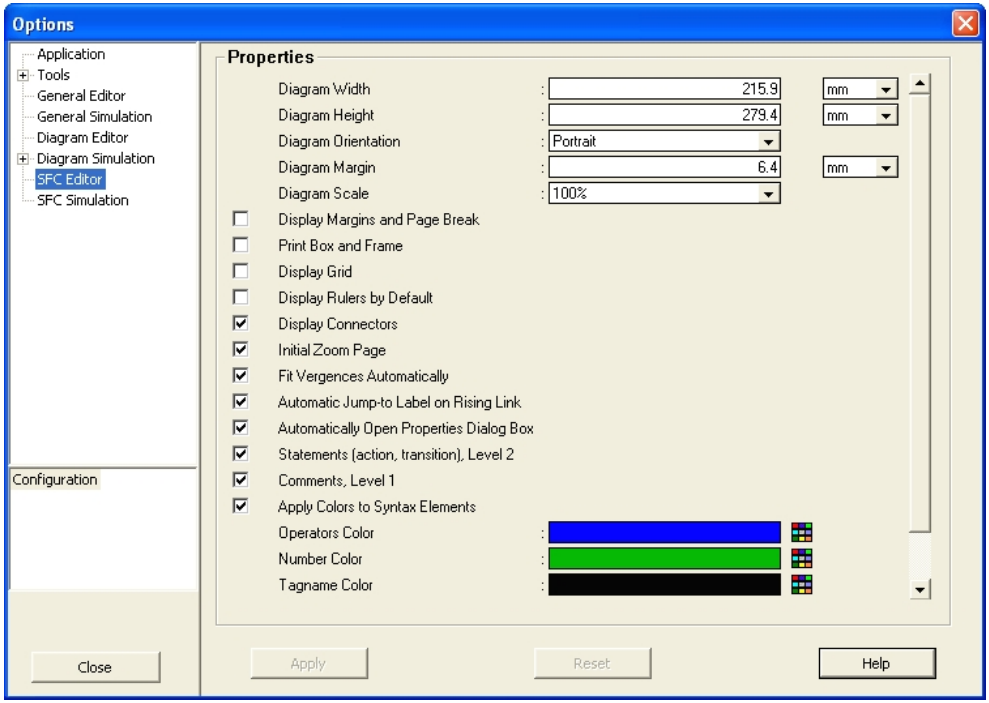


Figure 3-20: "SFC Editor" Branch in the "Tools" à "Options…" Dialog

Property	**Description**
Diagram Width	Specifies the height of the page.
Diagram Height	Specifies the width of the page.
Diagram Orientation	Specifies the orientation of the page (Portrait/Landscape).
Diagram Margin	Specifies the margin width of the page.
Diagram Scale	Specifies the scale of the page.
Display Margins and Page Break	Specifies whether to display margins and page breaks.

Property	Description
Print Box and Frame	Prints the box and frame.
Display Grid	Shows or hides the grid.
Display Rulers by Default	Shows or hides the rulers.
Display Connectors	Shows or hides the connectors.
Initial Zoom Page	Specifies whether an initial zoom is used for the page.
Fit Vergences Automatically	Specifies whether vergences are automatically adjusted or not.
Automatic Jump-to Label on Rising Link	Shows or hides the Jump-to label on the rising links.
Automatically Open Properties Dialog Box	Specifies whether the Properties Dialog Box is to open automatically.
Statements (action, transition), Level 2	Specifies whether or not to display level 2 statements.
Comments, Level 1	Specifies whether or not to display level 1 comments.
Apply Colors to Syntax Elements	Specifies whether or not to apply colors to syntax elements.
Operators Color	Specifies the color associated with operators.
Number Color	Specifies the colors associated with numbers.
Tagname Color	Specifies the colors associated with tagnames.

3.10.2 SFC Simulation

This dialog box allows the user to modify the default SFC simulation environment.

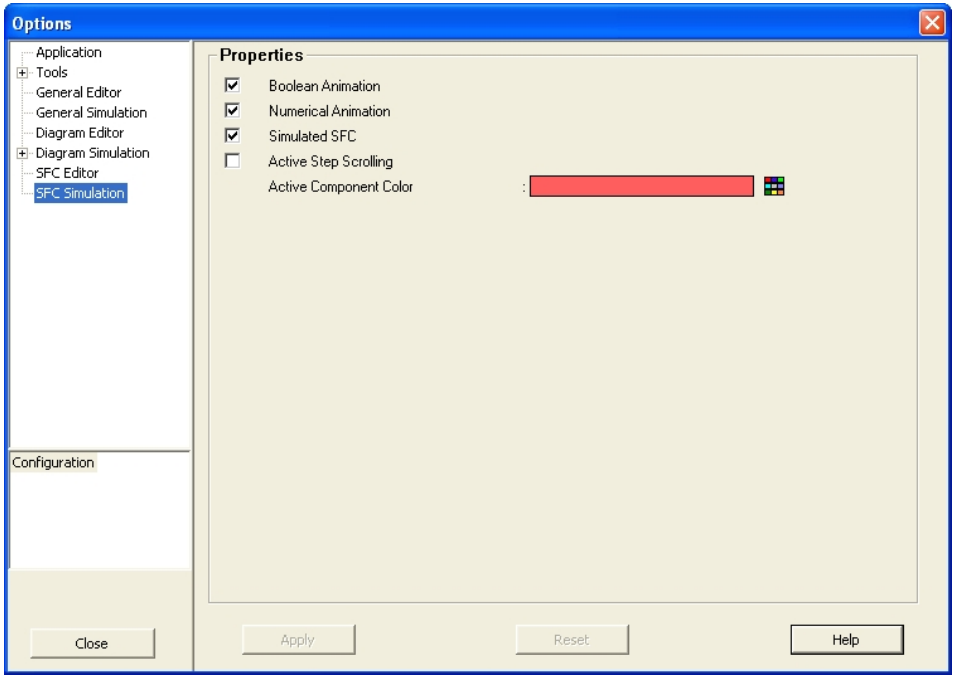


Figure 3-21: "SFC Simulation" Branch in the "Tools" à "Options…" Dialog

Property	Description
Boolean Animation	Specifies whether a Boolean animation is required.
Numerical Animation	Specifies whether a numerical animation is required.
Simulated SFC	Specifies if the graph is to be simulated or animated.
Active Step Scrolling	Specifies if the active step scrolling is desired.
Active Component Color	Specifies the colors used to indicate the active step or transition.

4 SFC Syntax

4.1 Introduction

A functioning SFC will treat one or more expressions, each composed of variables, constants, numbers and operators. The syntax defines the rules that must be respected for the identification of symbols and the writing of expressions.

This chapter presents a detailed description of syntax while using SFC components, symbols and processes. The following subjects are discussed:

- Identifier Nomenclature;
- Constants;
- Operators;
- Operator priority;
- Assignment;
- Syntax Verification;
- Timing;
- Managing SFC States and SFC Faults;
- Macro Steps
- Forcing Order.

4.2 Identifier Nomenclature

The identifier nomenclature allows a physical interpretation to be associated with every SFC symbol, be it a step, a transition or a variable. When one of these symbols is created, Automation Studio™ automatically generates a new name (except for variables), and assigns a unique number between 1 and 999.

Names are identifiers and must respect these basic rules:

- They are composed of standard alphanumeric characters: “a-z, A-Z, 0-9, _, +, -”;
- They always begin with a letter or number.
- They can be composed of multiple elements, each of these separated by a period (.).

Except for variables that are local to a SFC, variable tag names must be preceded by a period (.);

Variable	Type
variable	A variable associated with a project.
.variable	A variable associated with the current SFC.
.SFC1.variable	A variable associated with a SFC other than the current one, when both are in the same project.

- They are composed of 1 to 32 characters.
- They do not contain any empty characters (spaces).
- They do not contain any accented characters.
- The names are not case-sensitive. The names are converted to all-uppercase once Automation Studio™ recognizes them.

F If the SFC is used to control electro-pneumatic, ladder, electric or digital circuits make sure that the nomenclature of the components from those circuits follow these same rules.

4.2.1 Steps

When a new step is inserted, it is assigned a unique number by Automation Studio™. The number is used to refer clearly to this step in a given SFC.

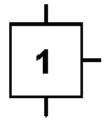


Figure 4-1: Inserting a Step

Moreover, two new variables are created for every newly inserted step. Those variables are available to the user:

- One variable represents the logical state of the step; *Xn.X*;
- One variable keeps track of the total time elapsed in milliseconds since the step was last activated. For example: *Xn.T*, *n* is the step number.

F The IEC 1131-03 international standard specifies that an activation period *Graph_Name.Xn.T/* starts with the rising step *Xn*. The length of the activation period is reset to zero only when the step again becomes active after it has been passive. The

elapsed time is an integer variable that indicates the number of milliseconds since the last activation.

F You must be careful when the same variable is used by a step and by a structured text. The actions associated with a step are evaluated in the sequential order of the SFC, while the structured text is updated with every simulation cycle.

4.2.2 Transition

When a new transition is inserted, it is assigned a unique number by Automation Studio™. The number is used to refer to this transition in a given SFC.



Figure 4-2: Inserting a Transition

4.2.3 Structured Text

A structured text behaves in the same way as an action, with the exception that the former is not associated with a step. A structured text is updated with every simulation cycle, after the step-associated actions have been executed.

As for steps and transitions, when a new structured text is inserted, it is assigned a unique number by Automation Studio™. The number is used to unambiguously refer to this structured text in a given SFC.

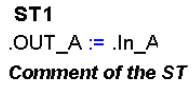


Figure 4-3: Inserting a Structured Text

By convention, the structured text is named *STn*, where *n* is the number of the structured text.

Structured texts are processed in the order that they are numbered after the normal steps of the SFC have been executed.

4.2.4 Comments

When a comment is inserted, the software will automatically assign to it a number within the 1 to 999 range. Each comment in the same SFC will have a unique number.

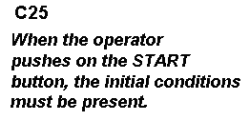


Figure 4-4: Inserting a Comment

By convention, a comment is named *Cn* where *n* is the number of the comment.

4.2.5 Reserved Keywords

The SFC workshop has reserved keywords, meaning that they cannot be used as identifiers.

- Names beginning with X are reserved for steps;
- Keywords "OR, XOR, AND, and NOT" are reserved for the Boolean operators;
- Keywords "F/" and "T/" are reserved for the forcing and timer operators;
- Keywords "MOD, ABS, SQRT, LN, LOG, EXP, SHL, SHR, ROL, ROR, INC, DEC, SIN, COS, TAN, ASIN, ACOS, ATAN, RAD_DEG, DEG_RAD, DEC_BCD, BCD_DEC, SEL, MIN, MAX, LIMIT, and MUX" are restricted to mathematical operators;
- Keyword "IF" is reserved for the conditional operator.

F When you insert actions and transitions, the expressions they contain will be displayed using a different colour if the expressions are not syntactically correct.

4.2.6 Variable Types

The variable type depends on the information it contains. Variable types are presented with the *Variables Manager* in the *Automation Studio™ User's Guide*. The following table lists the possible variable types that can be used for internal logic.

Variable type	Description
Boolean variable BOOLEAN	The value of a Boolean variable can either be 0 or 1, TRUE (ON), or FALSE (OFF).
Integer INTEGER	This variable's value is a whole number in the -2,147,483,648 to +2,147,483,647 range.

Variable type	Description
Real variable REAL	This variable's value is a real number as per IEC 559.

Time variable
TIME
	This variable's value is a time duration in the following format:
	T#3d_18h_4m_15s_23ms where:
	d = days;
	h = hours;
	s = seconds;
	ms = milliseconds.

4.2.7 Creating Variables

A new variable can be created with the help of the "New Variable" button found in the Properties dialog box of steps/transitions/structured texts or from the Variables Manager window. For more details on creating variables, refer to the *Variables Manager* in the *Automation Studio™ User's Guide*.

4.2.8 Creating External Links on Variables

It is possible to create external links (OPC) on variables with the use of the "External Links" button that appears in the Properties dialog boxes of steps/transitions/structured texts. For more details on creating external links, see *Variables manager* in the *Automation Studio™ User's Guide.*

4.3 Constants

Constants are fixed values that can be of the same type as variables.

4.4 Operators

An operator applies a function to one or more real and/or integer variables. This function returns a real or integer value that is typically assigned to a related variable.

4.4.1 Basic Arithmetic Expression Operators

Arithmetic operators are used to construct simple mathematical expressions involving variables. The returned valued is automatically converted to the expected type when

necessary.

The following table describes operator category, operand type and, if needed, a brief description and/or example.

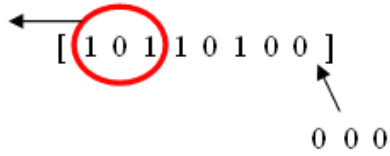
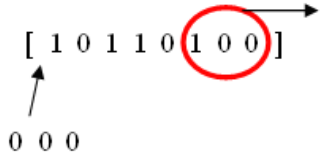
Operator	Description and example
+	Unary operation (positive value). E.g.: *Val1 := +25*
+	Addition E.g.: *Val1 := Val2 + Val3*
-	Unary operation (negative value). E.g.: *Val1 := -25*
-	Subtraction. E.g.: *Val1 := Val2 – Val3*
/	Division. E.g.: *Val1 := Val2 / Val3*
MOD	Remainder of the division operation, operands must be integers. E.g.: *Val1 := Val2 / Val3*
*	Multiplication. E.g.: *Val1 := Val2 * Val3*

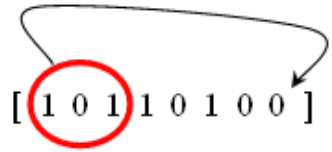
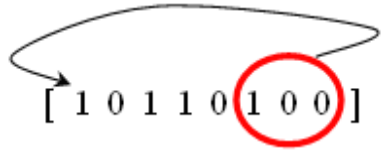
4.4.2 Advanced Mathematical Expression Operators

These arithmetic operators are used for complex mathematical expressions involving variables. The returned valued is automatically converted to the expected type when necessary.

The following table describes operator category, operand type and, if needed, a brief description and/or example.

Operator	Description and example
ABS	Absolute value, real or integer: ABS(.*Ls1*).
SQRT	Square root, real numbers only: SQRT(.*Ls1*).
LN	Napierian logarithm, real numbers only: LN(.*Ls1*).
LOG	Base 10 logarithm, real numbers only: LOG(.*Ls1*).
EXP	Natural exponent, real numbers only: EXP(.*Ls1*).
SIN	Sine, real numbers only: SIN(.*Ls1*).
COS	Cosine, real numbers only: COS(.*Ls1*).
TAN	Tangent, real numbers only: TAN(.*Ls1*).
ASIN	Arc sine, real numbers only: ASIN(.*Ls1*).
ACOS	Arc cosine, real numbers only: ACOS(.*Ls1*).
ATAN	Arc tangent, real numbers only: ATAN(.*Ls1*).
**	Exponent, real or integer: *.Ls1* ** *.Ls2*.
SHL	Shift left, integers only. .*Ls*1 := SHL(.*Ls*2,1), Left shift .*Ls*2 by 1 position The result is affected to .*Ls*1, .*Ls*2 keeps its original value. .*Ls*1 := SHL(.*Ls*2,.*Ls*3), Left shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*1, .*Ls*2 keeps its original value. .*Ls*2 := SHL(.*Ls*2,.*Ls*3) Left shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*2, .*Ls*2 takes on the final value.

Operator	Description and example
	Example : Enter the following integer: .A = [1 0 1 1 0 1 0 0], as shown.  Shifting 3 bits to the left gives: .B := SHL(.A, 3) = [1 0 1 0 0 0 0 0].
SHR	Shift right, integers only. .*Ls*1 := SHR(.*Ls*2,1), Right shift .*Ls*2 by 1 position The result is affected to .*Ls*1, .*Ls*2 keeps its original value. .*Ls*1 := SHR(.*Ls*2,.*Ls*3), Right shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*1, .*Ls*2 keeps its original value. .*Ls*2 := SHR(.*Ls*2,.*Ls*3) Right shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*2, .*Ls*2 takes on the final value. Example : Enter the following integer: .A = [1 0 1 1 0 1 0 0], as shown.  Shifting 3 bits to the right gives: .B := SHR(.A, 3) = [0 0 0 1 0 1 1 0].
ROL	Rotational left shift, and integers only. .*Ls*1 := ROL(.*Ls*2,1), Rotational left shift of .*Ls*2 by 1 position, .*Ls*2 is not modified. .*Ls*1 := ROL (.*Ls*2,.*Ls*3),

Operator	Description and example
	Rotational left shift of .*Ls*2 by .*Ls*3 positions. .*Ls*2 is not modified. .*Ls*2 := ROL(.*Ls*2,.*Ls*3) Rotational left shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*2. .*Ls*2 takes on the final value. Example : Enter the following integer: .A = [1 0 1 1 0 1 0 0], as shown.  A 3 bit rotational left shift gives: .B := ROL(.A, 3) = [1 0 1 0 0 1 0 1].
ROR	Rotational right shift, and integers only. .*Ls*1 := ROR(.*Ls*2,1), Rotational right shift of .*Ls*2 by 1 position, .*Ls*2 is not modified. .*Ls*1 := ROR (.*Ls*2,.*Ls*3), Rotational right shift of .*Ls*2 by .*Ls*3 positions, .*Ls*2 is not modified. .*Ls*2 := ROR(.*Ls*2,.*Ls*3) Rotational right shift of .*Ls*2 by .*Ls*3 positions The result is affected to .*Ls*2, .*Ls*2 takes on the final value. Example : Enter the following integer: .A = [1 0 1 1 0 1 0 0], as shown.  A 3 bit rotational left shift gives:, .B := SHR(.A, 3) = [1 0 0 1 0 1 1 0].
INC	Incrementation of real or integer values.

Operator	Description and example
	.*Ls*1:= INC(.*Ls*2, 1), The value of .*Ls*2 incremented by 1 the result is affected to .*Ls*1. INC(.*Ls*2,.*Ls*3) The value of .*Ls*2 is incremented by .*Ls*3.
DEC	Decrementation of real or integer values. .*Ls*1:= DEC(.*Ls*2, 1), The value of .*Ls*2 decremented by 1 the result is affected to .*Ls*1. DEC(.*Ls*2,.*Ls*3) The value of .*Ls*2 is decremented by .*Ls*3.
SEL	Binary selection between Booleans, reals or integers. .*Ls*1 :=SEL(G, In1, In2) Binary selection, if G=1, output = In2 The selection indicator G must be a Boolean, if G=0 output = In1, if G=1 output = In2
MIN	Minimum of multiple values, real or integer. .*Ls*1:= MIN(In1, In2, …, In16) Minimum value of a maximum 16 variables.
MAX	Maximum of multiple values, real or integer. .*Ls*1:= MAX(In1, In2, …, In16) Maximum value of a maximum 16 variables.
LIMIT	Value limiter, real or integer. .*Ls*1:=LIMIT(IN, MIN, MAX) Limiter, Output = MIN threshold if input is lesser than MIN threshold, Output = MAX threshold if input is greater than MAX threshold, Output = IN if input value is between MIN and MAX thresholds. *.Ls1* :=LIMIT(*.Ls2*, *.Ls3*,*.Ls4*),

Operator	Description and example
MUX	Multiplexing between multiple inputs, Booleans, reals or integers. .*Ls*1:= MUX(K, In1, In2, …, In16) Multiplexer The output is the input value selected by K, K must be an integer in the range of 0-16. If K=0, output is 0 (false for Booleans). Instruction MUX(K, In1, In2, …, In16): If K is >16 or <0, the instruction is not executed and a message will appear in the Messages Window.
BCD_TO_DEC	Conversion from BCD to decimal, integers only. .*Ls*1:= BCD_TO_DEC(.*Ls*2)
DEC_TO_BCD	Conversion from decimal to BCD, integers only. .*Ls*1:= DEC_TO_BCD(.*Ls*2)
RAD_TO_DEG	Conversion from radians to degrees, integers only. .*Ls*1:= RAD_TO_DEG(.*Ls*2)
DEG_TO_RAD	Conversion from degrees to radians, integers only. .*Ls*1:= DEG_TO_RAD(.*Ls*2)

4.4.3 Boolean Expression Operators

These operators are used to evaluate logical expressions involving variables. The returned value is a Boolean.

The following table describes these operators.

Operator	Description
OR	Logical OR between Boolean values. E.g.: .*Ls*1 OR .*Ls*2
XOR	Exclusive logical OR between Boolean values. E.g.: .*Ls*1 XOR .*Ls*2
AND	Logical AND between Boolean values. E.g.: .*Ls*1 AND .*Ls*2
NOT	Negation of a Boolean value. E.g.: NOT .*Ls*1

4.4.4 Edge Operators

An expression involving an edge operator is a Boolean expression. It is triggered by a change in state.

An edge is a special type of operator, that indicates when a Boolean variable or Boolean expression changes its value.

There are two types of operators: rising edge operator and falling edge operator. They are defined in the following table:

Operator	Description
^B or R_TRIG(B)	Rising edge. Rising edge operator. The edge operator becomes TRUE when the operand changes from FALSE to TRUE. Example: A IF ^B. A becomes TRUE when B changes from FALSE to TRUE.

Operator	Description
! or F_TRIG(B)	Falling edge. Falling edge operator. The edge operator becomes TRUE when the operand changes from TRUE to FALSE. Example: A IF !B. A becomes TRUE when B changes from TRUE to FALSE.

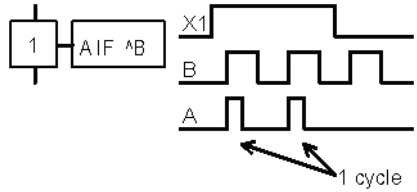


Figure 4-5: Timing Diagram of an Action using a Rising Edge Operator

4.4.5 Relational Operators

These operators are used to compare variables and/or numerical values (numbers, constants).

The result is a Boolean value. If the comparison is true the output is TRUE and FALSE if the comparison is false.

Operator	Description
>	Expression A > B is true if the value of A is greater than the value of B.
>=	Expression A >= B is true if the value of A is greater than or equal to the value of B.
=	Expression A = B is true if the value of A is equal to the value of B.
<=	Expression A <= B is true if the value of A is lesser than or equal to the value of B.
<	Expression A < B is true if the value of A is lesser than the value of B.

Operator	Description
<>	Expression A <> B is true if the value of A is different from the value of B.

4.5 Operator Priority

The priority of operators influences the way that expressions are evaluated.

The following table lists the operators by order of priority. The operator on top has the highest level of priority while the operator at the bottom has the lowest. For operators on the same row, the operator on the left has more priority than the operator on the right.

Operator	Priority
()	Parentheses
+, -, NOT, ^, !	Unitarian Operators
*, /	Multiplication Operators
+, -	Additive Operators
<, <=, >, >=	Relative Operators
=? <>	Relative Operators
XOR	Additive Operators
AND	Additive Operators
OR	Additive Operators

The only way to override these priorities is by using parenthesis. In that case, the expression that is within the most parenthesis has the highest priority.

4.6 Assignment Operators

Assignment operators define the types of action and their effects. They can be applied to integers, reals, Booleans. They are used in actions and structured texts. The following section describes the different assignment operators and their use.

The execution of an action is conditioned by the active state of the step and the SFC, unless the unconditional option is checked for the step. In this case, the execution of the action is dependant only of the active state of the SFC.

4.6.1 Continuous Boolean Assignment

The Continuous Boolean Assignment (or unconditional action in a step) returns the value TRUE to the variable as long as the assignment continues. There are no assignment operators. The variable becomes TRUE when the step is activated then becomes FALSE when it is deactivated.

The syntax of a continuous Boolean action is the same as the name or tag name of the variable. In this case, you must not use an assignment operator.

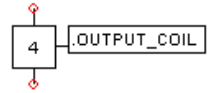


Figure 4–6: Continuous Boolean Assignment in an Action

In this example:

- COIL is a Boolean variable that is global to the project;
- When step 4 is activated, COIL becomes TRUE and will retain that value until step 4 is deactivated;
- When step 4 is deactivated, COIL returns to FALSE;

4.6.2 Numeric Assignment

The Numeric Assignment Operator assigns a value (variable, constant or the result of a calculation) to a numeric value. The numeric assignment operator sign is " :=".

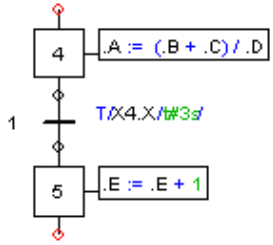


Figure 4-7: Numeric Assignment

In this example:

- Variables A, B, C, D, and E are real numerical variables;
- When step 4 is active, variable A is assigned the value of the expression (B + C) / D;
- Variable A keeps the last value of (B + C) / D when step 4 is deactivated;
- When step 5 is active, the variable E is incremented by 1.

4.6.3 Conditional Boolean Assignment

The Conditional Boolean Assignment makes it possible to assign the state of a variable or of an expression to a Boolean variable if a certain condition is met. In the contrary case, FALSE is assigned to the Boolean variable. The conditional assignment operator's sign is: “IF”.

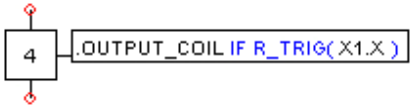


Figure 4-8: Conditional Boolean assignment

In this example:

- When step 4 is active the state of the Boolean variable COIL is the same as the state of the R_TRIG(X1.X) expression;
- When step 4 is inactive, the Boolean variable COIL is FALSE.

4.6.4 Conditional Numerical Assignment

This type of assignment is done when the specified condition is TRUE. The conditional numerical assignment operators are: “IF” and “:=”.

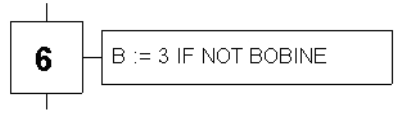


Figure 4-9: Conditional numerical assignment

In this example:

- When step 6 is active, variable B equals 3 if the expression NOT COIL is TRUE;
- When step 6 is inactive, B keeps its last value.

4.6.5 Stored Boolean Assignment

The stored Boolean assignment makes it possible to assign to a variable a state of TRUE or FALSE, a state that it retains (stored). To assign the state of TRUE or FALSE to a variable, the operators of assignment are respectively: ":= 1" or ":= 0".

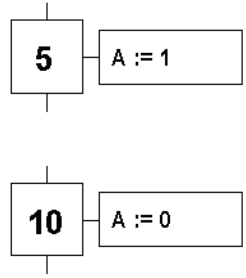


Figure 4-10: Stored assignment

In this example:

- Variable A equals TRUE when step 5 is active and keeps this value after the deactivation of step 5;
- Variable A equals FALSE when step 10 is active and keeps this value when step 10 is deactivated.

4.7 Syntax Analysis

The SFC Editor uses syntax highlighting. This makes it easy for the user to identify syntax errors while typing actions, transitions or structured texts.

For syntax analysis to be enabled, make sure the option "Apply Colors to Syntax Elements" is activated in the SFC Editor branch, of the "Options" dialog box. See *3.10 SFC Environment*

for more information.

4.8 Timing

F When timers are used, their time information is measured using the simulated time. The simulated time is shown in the status bar when in the simulation mode. For more information, refer to the *Automation Studio™ User's Guide.*.

There are four (4) types of timers:

Type	Description
(T/, TON/);	Timer On-delay.
(TOF/);	Timer Off-delay.
(TP/);	Pulse delay timer
(Xn.T).	Cumulative time since last timer activation.

Timers are Boolean operators that take time into consideration (delay, wait state, etc.). They are written in the following forms:

- /Xn.X/T#a/;
- TON/Xn.X/T#a/;
- TOF/ Xn.X/T#a/;
- TP/ Xn.X/T#a/.

where:

Characters	Description
T/, TON/, TOF/, TP/	Timing operator.
Xn.X	Identifies the input variable, the step or the variable that will trigger the timer.

Characters	Description	
T#a	Length of duration time variable. ex. : T#30s where :	
	day	j
	hour	h
	minute	m
	second	s
	millisecond	ms

Timers can be used to define an action that is delayed, or whose duration is limited.

Timer On-delay

The timing operation *TON/Xn.X/T#a* evaluates to the Boolean value "1" only when both the following are true:

- The duration defined by *T#a* is over;
- The variable *Xn.X* is active.

The accumulated time is set to zero whenever the variable is inactive. When the variable is active, the cumulated time increases from zero up to *T#a*,

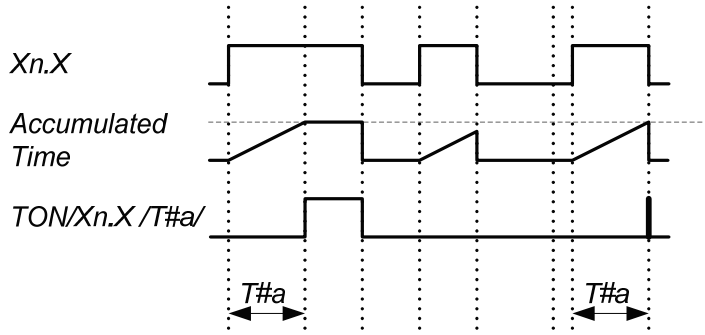


Figure 4-11: Timer On-delay

Delayed Actions

The timing operation *TOF/Xn.X/T#a/* evaluates to the Boolean value "1" only when either of the following is true:

- The variable *Xn.X* is activated.
- The duration defined by *T#a* is not over;

The accumulated time increases from zero when the variable becomes inactive, up to *T#a*. The accumulated time is reset when the variable is activated.

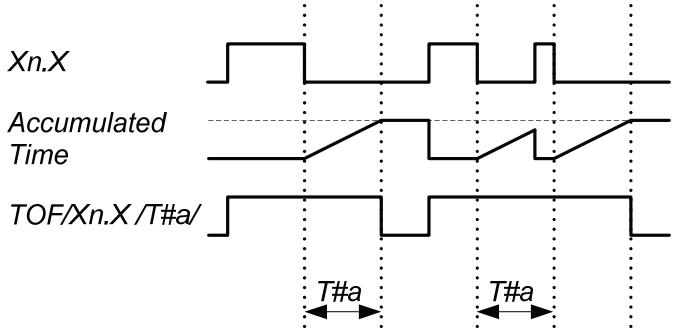


Figure 4-12: Timer Off-delay

Time-Limited Actions

The timing operation *TP/ Xn.X/T#a/* evaluates to the Boolean value "1" only when the duration defined by *T#a* is not complete.

The accumulated time increases from zero when the variable become active, *T#a*. The accumulated time is then reset only when the variable is deactivated.

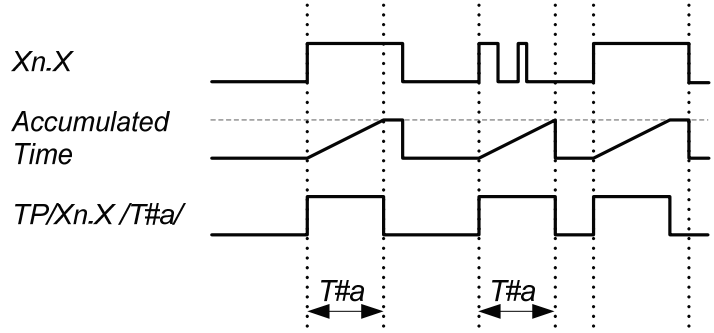


Figure 4-13: Pulse delay timer

4.9 Managing SFC States and SFC Faults

A SFC can be interpreted as a black box that receives an input and generates an output.

Variables and commands are the inputs given to a SFC. The SFC's output is a reflection of the new configuration of its internal states, and new values for its variables.

Commands and states are part of the SFC.

The state variables that can be modified are :

Internal SFC Variable	Description
INIT	SFC initialisation.
RUN	Authorisation to run the SFC and execute the actions associated with the steps.
ESC	External condition that cancels the SFC's run.
RUNNING	State of the SFC being executed.
FORCED	Forced state of the SFC.
GEN_FAULT	General SFC fault : indicates a fault in the SFC, or the pressing of ESC.

These states and faults can be used to manage the "RUN" execution mode, the initialisation (INIT) mode and/or the forcing (FORCED) of the SFC. If this type of variable is used, a mechanism should be used to warn, then to receive acknowledgement from the user. For example, a warning light can be used to signal a fault, and a push-button can be used to acknowledge.

Fault variables cannot be modified. They are listed in the following table:

Internal SFC Variable	Description
STEP_MAX_ACK	Acknowledgement command for the maximum step duration fault.
STEP_MIN_ACK	Acknowledgement command for the minimum step duration fault.
SEQ_MAX_ACK	Acknowledgement command for the maximum sequence duration fault.
SEQ_MIN_ACK	Acknowledgement command for the minimum sequence duration fault.
STEP_MAX_FAULT	Maximum step duration fault (only one fault for all steps under surveillance).
STEP_MIN_FAULT	Minimum step duration fault (only one fault for all steps under surveillance).
SEQ_MAX_FAULT	Maximum sequence duration fault.
SEQ_MIN_FAULT	Minimum sequence duration fault.

4.10 Macro Steps

A macro step is a section of the SFC represented condensed in a single step. For each macro step there is a macro step SFC that explodes the source macro step. The name of the SFC is inscribed in the action box on the right hand side of the macro step.

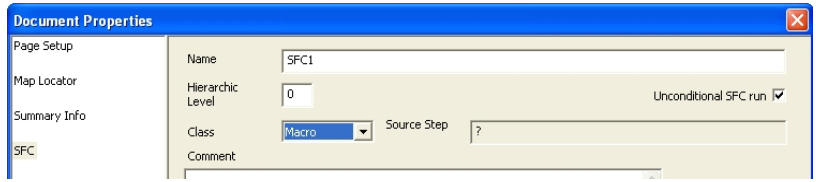


Figure 4-14: Defining a SFC as a macro

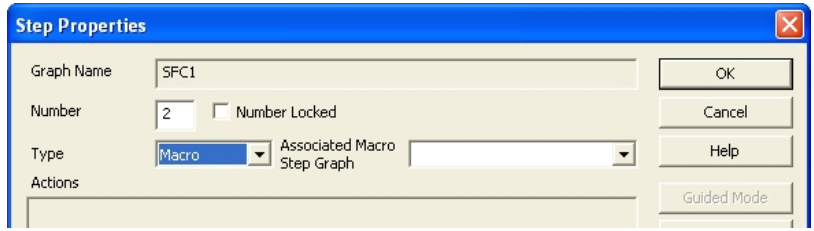


Figure 4-15: Defining a macro step

The in step of the macro step SFC and the parent macro step SFC are activated simultaneously. It is in fact an implicit AND divergence.

The out step of the macro step SFC is deactivated simultaneously with the parent macro step SFC. This happens when the transition that follows the parent macro step SFC is fired. It is in fact an implicit AND convergence.

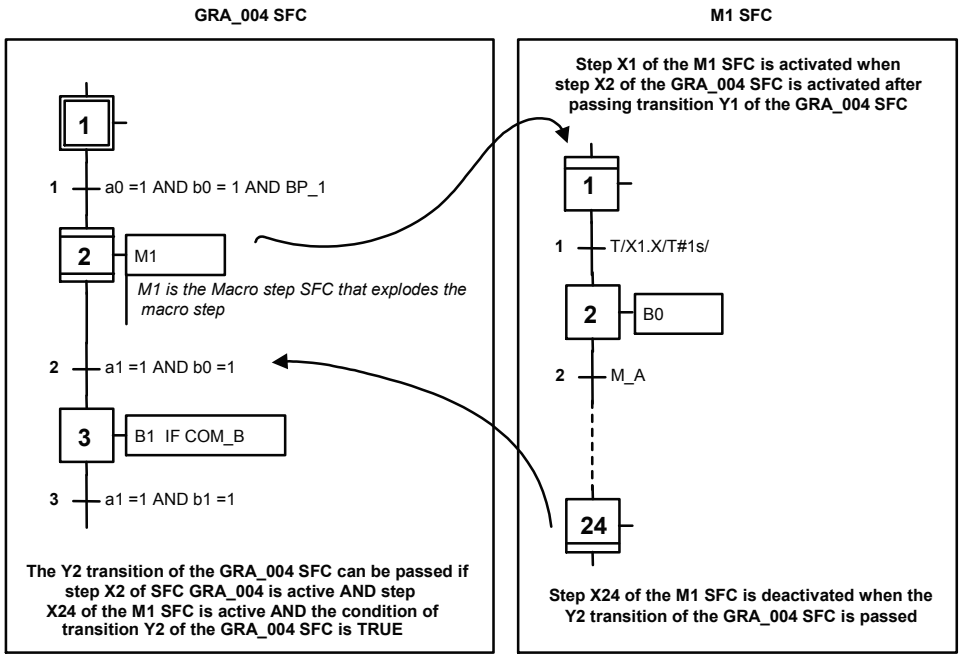


Figure 4-16: Using Macro Steps

4.11 Hierarchical Forcing

Forcing statements permit the modification of the SFC state. They are used to impose a certain situation on an SFC. The forcing statement will run as long as its activation condition is TRUE.

F During simulation, it is possible to force the activation of a step or the passing of a transition without respecting the SFC syntax. This function must not be confused with hierarchical forcing.

The general form for forcing is:

F/ SFC_Name.FORCED:(.SFCn.Xn.FORCED…),

where,

F/ Forcing operator.

.SFCn.FORCED	Name of the SFC that will be forced.
:(SFCn.Xm.FORCED,…)	List of steps that will be forced, this corresponds to the desired SFC situation. One or more steps can be specified for one SFC and must be separated by a comma. If no step is specified (), all steps of the SFC are deactivated.

During forcing, specified steps are activated and all others deactivated.

4.11.1 Rules on Hierarchical Forcing

- An SFC cannot force itself;
- An SFC cannot force another SFC on the same hierarchical level;
- An SFC cannot force another SFC on a higher hierarchical level;
- Two SFCs on the same hierarchical level cannot force the same SFC;
- If an SFC if forced by two SFCs of higher and different hierarchical levels, all the specified steps in both instructions are forced. The last active instruction will determine the state of the SFC when the instruction is deactivated;
- An SFC can be forced even if it isn't running. (see *4.9 Managing SFC States and SFC Faults*);

The following figures depict legal and illegal forcing actions:

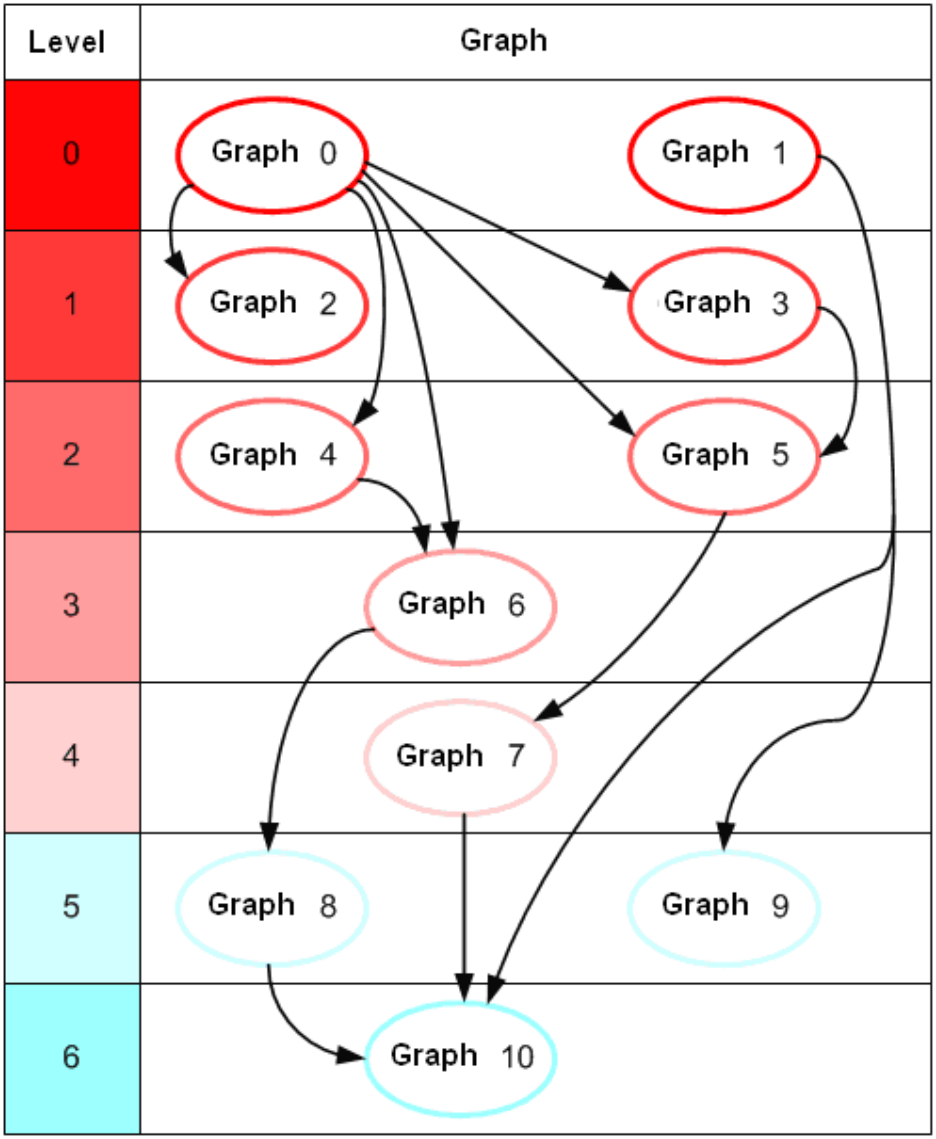


Figure 4-17: Hierarchical Forcings allowed

5 Predefined SFC Variables

Some variables are automatically created with a new SFC, and when new components are inserted.

These variables are either readable and/or writable, and are available in both the simulation and edition modes.

The Boolean variables that are automatically created with a SFC are listed below :

Variable	Description
ESC	External security condition.
FORCED	Signals that a SFC is forced if any of its steps is forced. This step must not depend on RUN. A SFC cannot force itself, nor with a SFC at a hierarchical level equal to or greater than itself. Two or more SFCs with the same hierarchical level cannot force an identical SFC. If a SFC is forced by two or more SFCs, the forcing by the SFC at the highest level of the hierarchy has priority.

Variable	Description
GEN_FAULT	General SFC fault. If a fault is present but not acknowledged, or if the external security condition is present, GEN_FAULT is automatically set to TRUE according to the following equation : GEN_FAULT := SEQ_MIN_FAULT OR SEQ_MAX_FAULT OR STEP_MIN_FAULT OR STEP_MAX_FAULT OR ESC A General SFC fault does not trigger a fault. The user can use these states to RUN, INIT or FORCE the SFC. For example : .SFC1.ESC IF EMERGENCY_STOP .SFC1.RUN := NOT .SFC1.GEN_FAULT .SFC1.INIT := NOT .SCF1.GEN_FAULT
INIT	The initialization command : Activates the initial steps, and inactivates the others. This variable is set to TRUE by the simulator on the first simulation cycle. Except for the SFC at hierarchical level zero, the INIT command can be triggered by an action, or by another SFC. A SFC cannot initialize itself.
RUN	The RUN command authorizes the evolution of the SFC and the execution of the actions associated with its steps. Except for SFC at hierarchical level zero, the RUN command can be triggered by an action, or by another SFC. A SFC cannot run itself.
RUNNING	State of a SFC being executed.

Variable	Description
SEQ_MAX_ACK	Acknowledgement command for the maximum sequence duration fault.
SEQ_MAX__ACK_MEM	Acknowledgement memorization of the maximum sequence duration fault.
SEQ_MAX_FAULT	Maximum sequence duration fault.
SEQ_MIN_ ACK	Acknowledgement command for the minimum sequence duration fault.
SEQ_MIN_ ACK_MEM	Acknowledgement memory of the minimum sequence duration fault.
SEQ_MIN_FAULT	Minimum sequence duration fault.
STEP_MAX_ ACK	Acknowledgement command for the maximum step duration fault.
STEP_MAX_ ACK_MEM	Acknowledgement memorization of the maximum step duration fault.
STEP_MAX_ FAULT	Maximum step duration fault (only one fault for all steps under surveillance).
STEP_MIN_ ACK	Acknowledgement command for the minimum step duration fault.
STEP_MIN_ ACK_MEM	Acknowledgement memorization of the minimum step duration fault.
STEP_MIN_ FAULT	Minimum step duration fault (only one fault for all steps under surveillance).

F Variables associated with a SFC are denoted by:

.nameOfSFC.FORCED

The variables (n is the step number.) automatically created with a component are listed below :

Variable	Type	Description
Xn.X	Boolean	State of the step.
Xn.T	Integer	Accumulated activation time of the step.
Xn.FORCED	Boolean	Forced state of the step.

F Variables created with a component are denoted by *.nameOfSFC.Xn.X.*

6 Creation of a First SFC

This example will guide you through every step necessary to build your first SFC.

6.1 Operative Part

Before starting to build your first SFC, please open the file *SFC_PNEU_EN..pr5* from the Pneumatics Workshop, or the file *SFC_HYDRO_EN.pr5* from the Hydraulics Workshop. These files contain a pneumatic (or hydraulic) circuit, together with a control SFC. The files can be found in the *Demos_and_Tutorials* directory, and can be used as a basis to build other SFCs.

Create a new project in which you will build a circuit identical to the one in the file. You can also copy and paste the circuit into your new document. Give the components the same names as in the original file.

F For clarity, we will only use the pneumatic circuit in this guide.

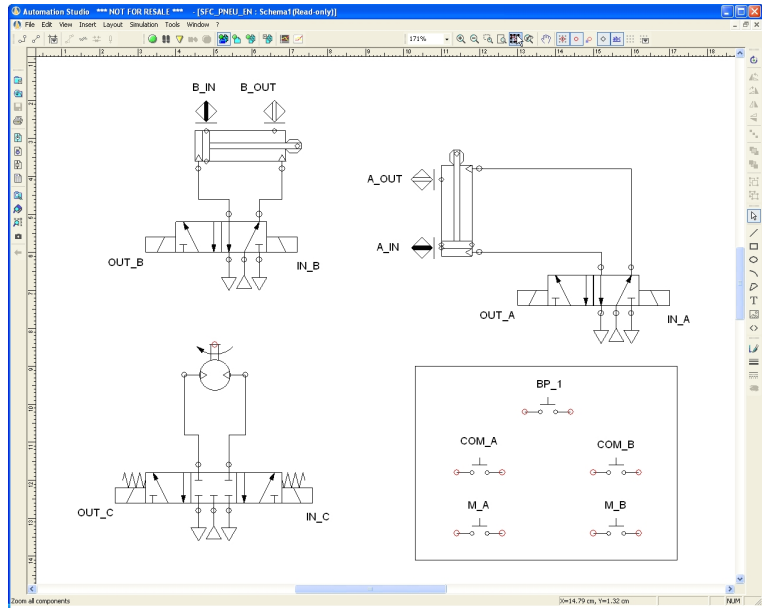


Figure 6-1: Sample pneumatic/electric circuit

F Tagnames for the solenoids are assigned after the internal SFC variables have been created.

The circuit moves a mass from level 1 to level 2, and places it in either the left or right bin, according to its weight. The whole process is composed of three steps:

1. The extension of the piston rod in cylinder A moves the mass up;

2. The extension of the piston rod in cylinder B pushes the mass on level 2;

3. A clockwise or counterclockwise rotation of the motor puts the mass in either bin.

The control steps are:

1. Start; *BP_1*;

2. Activate cylinder A; *COM_A*;

3. Activate Cylinder B; *COM_B*;

4. Push mass into the left or right bin; *M_A* or *M_B*.

It is the SFC's responsibility to generate these commands and to control the process.

6.2 Creating a new SFC

The following steps explain how to use the various menus and dialog boxes to create a new SFC. More details will be given later.

1. To create a new SFC;

- Choose "File" on the menu bar, then "New" à "Sequential Function Chart";

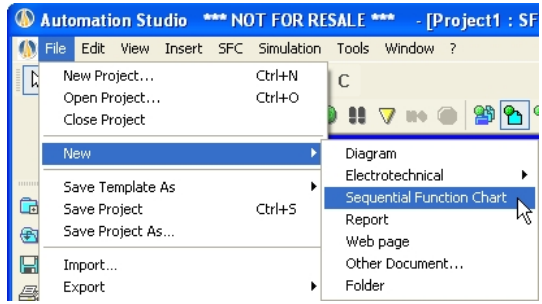


Figure 6-2: Creating a new SFC

OR

- Click on the "New Sequential Function Chart" icon ( ) on the toolbar;

  The "Sequential Function Chart Templates" dialog box will open (*Figure 6-3: "Sequential Function Chart Templates" dialog box*).

2. Choose the icon marked "None", then click on "OK";

   The "Document Properties" dialog box will open (*Figure 6-4: "Document Properties" dialog box for a new SFC*).

3. Click on "Close". You can start creating your new SFC. (*Figure 6-5: SFC creation, main window*).

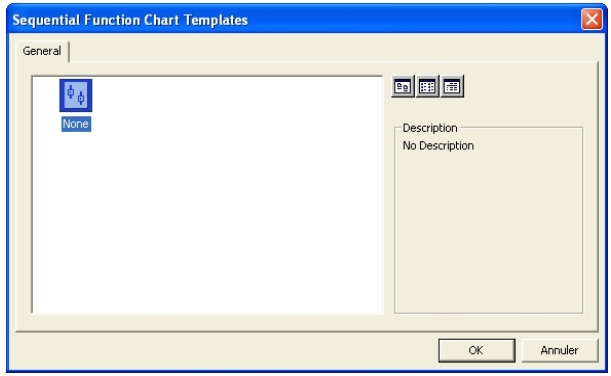


Figure 6-3: "Sequential Function Chart Templates" dialog box

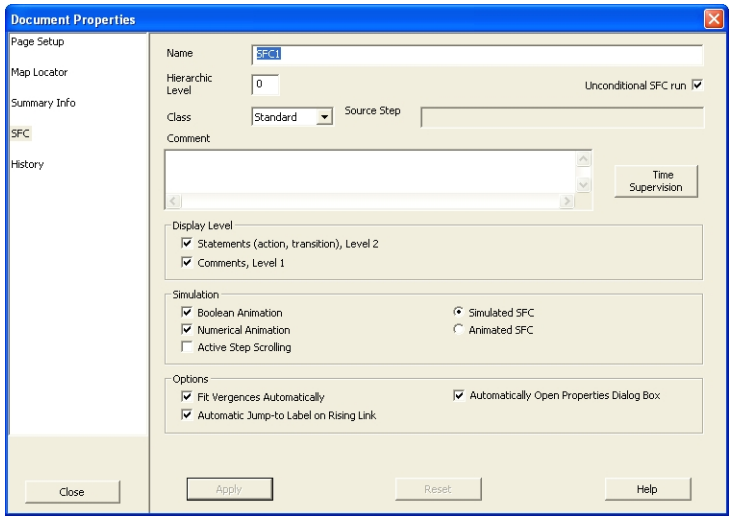


Figure 6-4: "Document Properties" dialog box for a new SFC

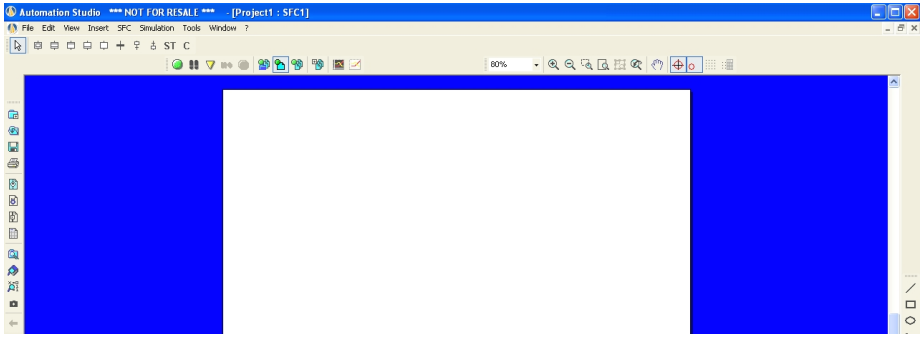


Figure 6-5: SFC creation, main window

6.3 Editing the SFC

The following sections will guide you through the steps to build the SFC shown in the figure below. Note that no components from the library can be dropped on the SFC document.

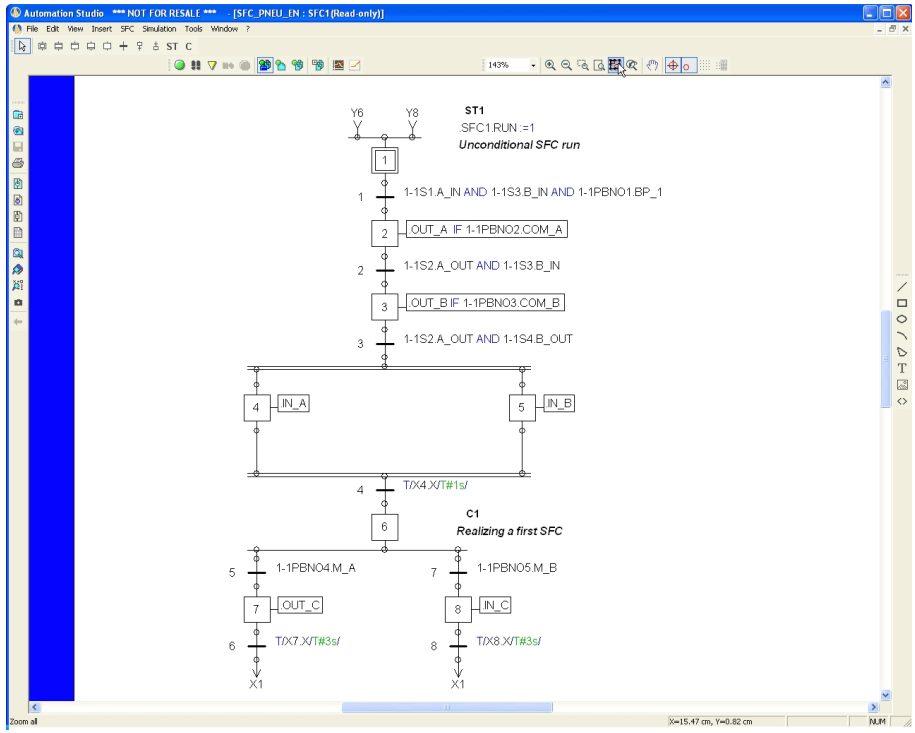


Figure 6-6: SFC

F A component from the SFC Workshop can only be connected to a component from the same workshop.

6.3.1 Initial Step

1. In the SFC toolbar, click on "Insert Initial Step" ( );

2. Move the mouse pointer on the SFC document where you wish to place the new step, and left-click;

   The "Step Properties" dialog box opens

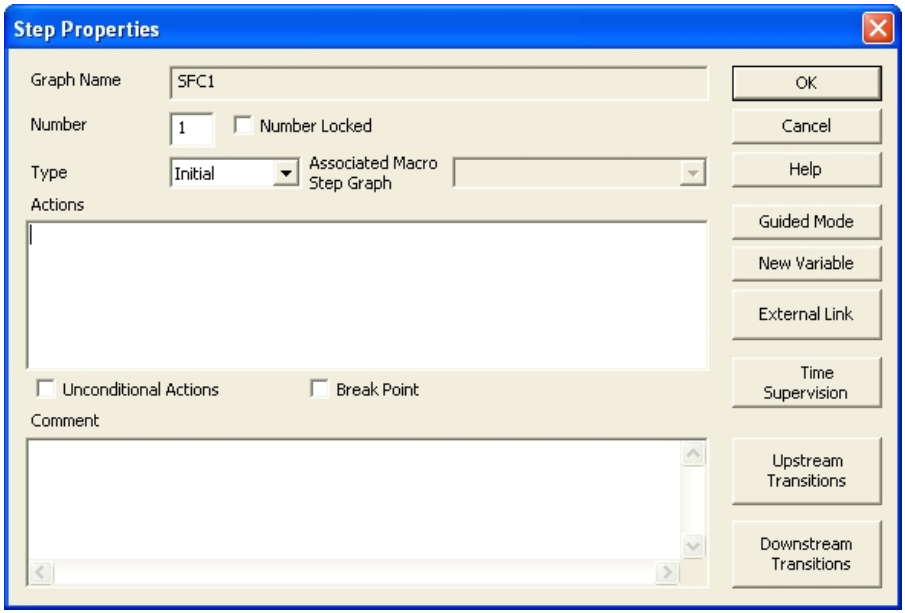


Figure 6-7: "Step Properties" dialog box for the initial step

For this step:

3. Select "Initial" in the drop-down list;

4. Click "OK".



Figure 6-8: Initial Step

6.3.2 First Transition

1. In the SFC toolbar, click on "Insert Transition Step" ();

2. Move the mouse pointer to the output connector of step 1;

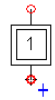


Figure 6-9: Inserting a Transition

Left-click. The "Transition Properties" dialog box opens.

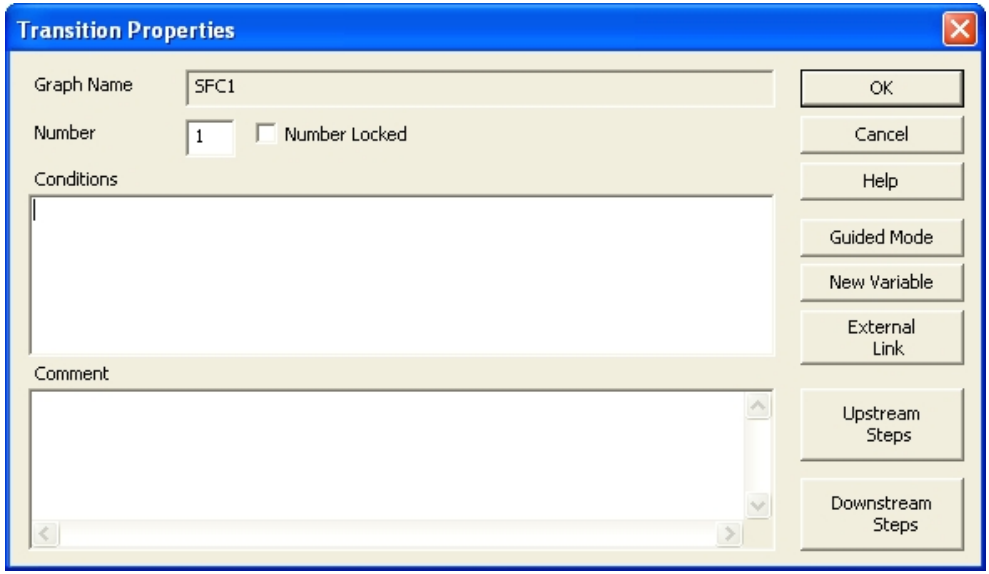


Figure 6-10: "Transition Properties" dialog box, for first transition

For this transition :

3. Ensure that the Number is 1;

4. Click the "Guided Mode" button.

   The "Transition Guided Mode" dialog box opens.

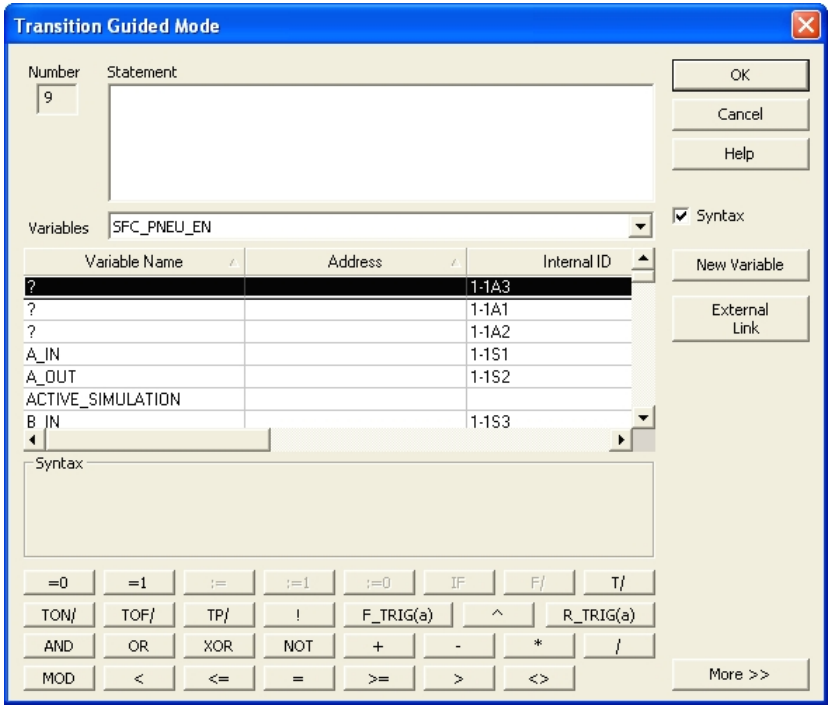


Figure 6-11: "Transition Guided Mode" dialog box

5. Write the equations for the transition :

F To avoid erasing the equations already in the "Statement" field with a new instruction, make sure the cursor is correctly positioned on the field.

a. Find the variable *A_IN* in the list of variables;
b. Double-click on the variable to insert it in the "Statement" field;
c. Click the "AND" button;
d. Find the variable *B_IN* in the list of variables;
e. Double-click on the variable;
f. Click the "AND" button;
g. Find the variable *BP_1*;
h. Double-click on the variable;

The equation is now complete. It should be shown in the "Statement" field.

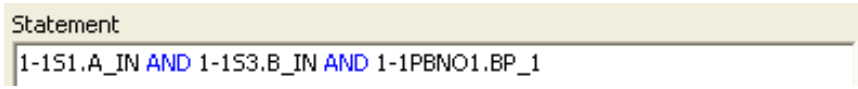


Figure 6-12: Equation for the first transition

i. Click on "OK"

The "Transition Guided Mode" dialog box closes, and we get back to the "Transition Properties" dialog box. The new equation is displayed in the "Conditions" field..

6. Click on "OK".

The SFC now has two components.

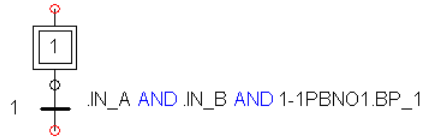


Figure 6-13: SFC with two components

6.3.3 First Action

1. In the SFC toolbar, click on "Insert Standard Step" ( );
2. Move the mouse pointer to the output connector of transition 1;

   Left-click. The "Step Properties" dialog box opens..
3. Click the "Guided Mode" button.

   The "Step Guided Mode" dialog box opens
4. Write the equations for the step ;

   a. Create a new Boolean variable *OUT_A* ;

      i. Click the "New Variable" button;

      The "Add a variable" dialog box opens.

      ii. Enter **OUT_A** in the "Tagname" field;

iii. Click on "OK".

The new variable appears in the "Statement" field.

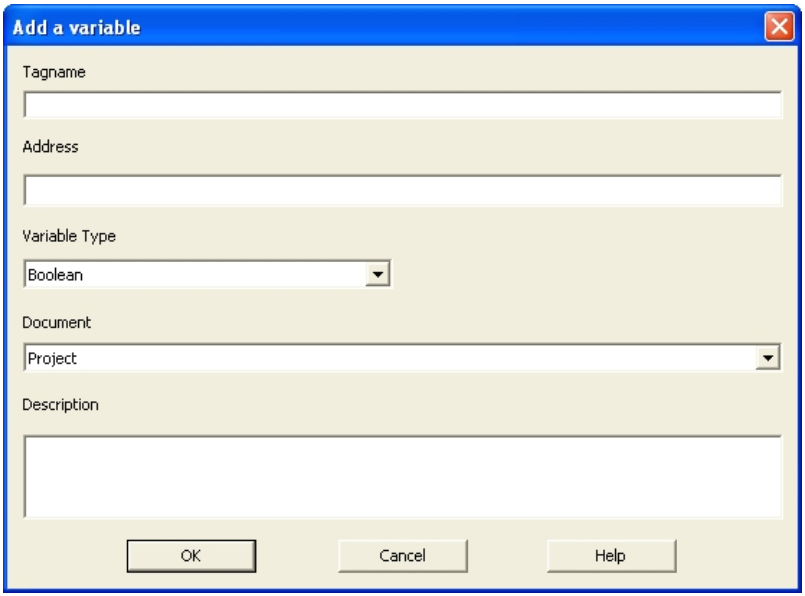


Figure 6-14: "Add a variable" dialog box

b. Click the "IF" button;

c. Find the variable *COM_A* in the list of variables;

d. Double-click on the variable;

The equation is now complete. It should be shown in the "Statement" field.



Figure 6-15: Equation for the first standard step

e. Click on "OK"

The "Step Guided Mode" dialog box closes, and we get back to the "Step Properties" dialog box. The new equation is displayed in the "Actions" field..

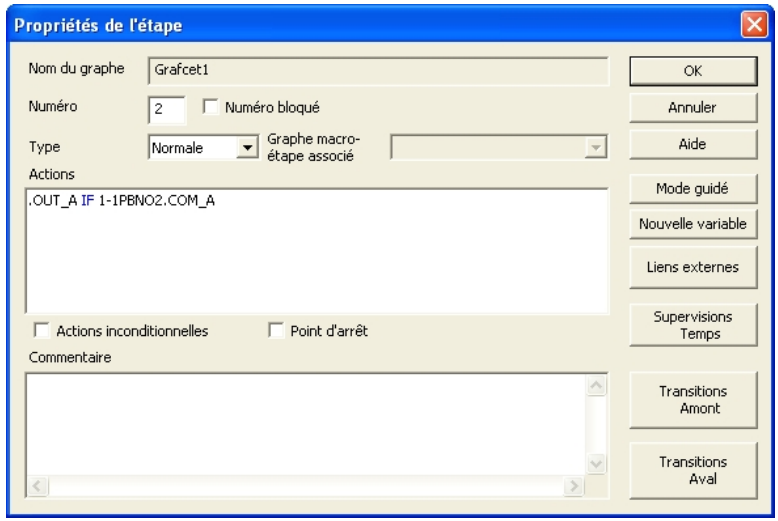


Figure 6-16: "Step Properties" dialog box with the new actions

F After a symbol such as a step or a transition has been inserted in the SFC, it is possible to modify its properties. To do this, just double-click on the symbol to open the properties dialog box.

The SFC now has three components.

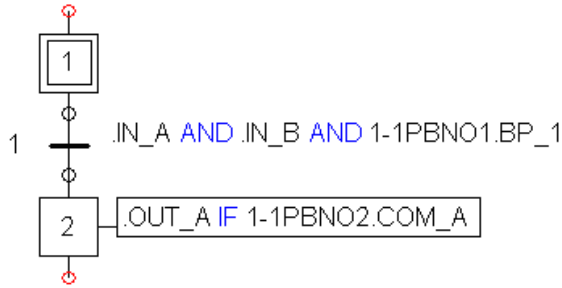


Figure 6-17: SFC with three components

6.3.4 Inserting Other Steps and Transitions

Using the steps described in the two previous sections, insert all the transitions and steps into the SFC.

Some components should be linked by a convergence or divergence. For now, leave those

components free. You should obtain the following SFC :

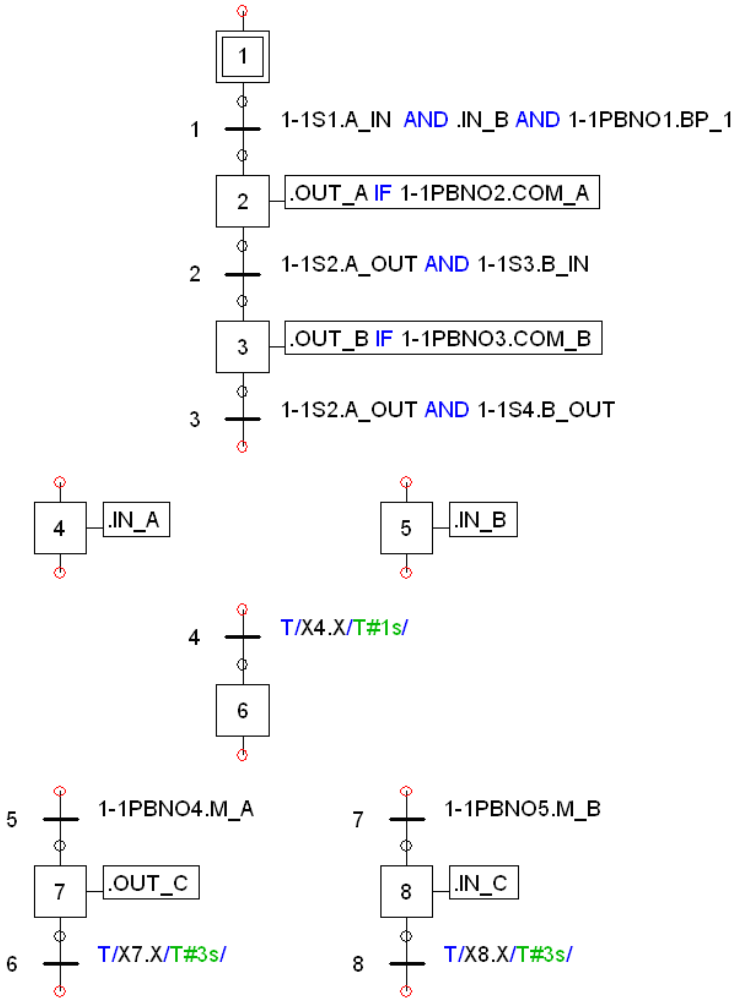


Figure 6-18: Complete SFC, without vergences

6.3.5 Inserting Links

When a component is inserted on the connector of another component, the new component is automatically linked with the first.

If a component is not inserted on a connector, you must manually link it with another

component.

To create a link between two components :

1. In the SFC toolbar, click on "Link" ( ); 

2. Place the pointer on the component's connector;

   A circle around a connection point represents a connector. A connector that is correctly linked is displayed using a different color. The mouse pointer changes its appearance when it hovers on a connector to indicate that the connection tool is available. To know more, refer to the *Layout of Links* section in the *User's Guide*.

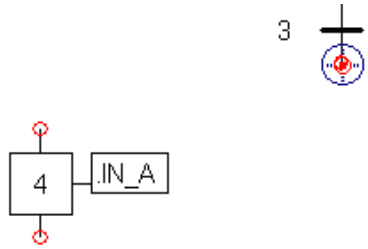


Figure 6-19: Layout of Links - First connector

3. Click the connector on the initial component;

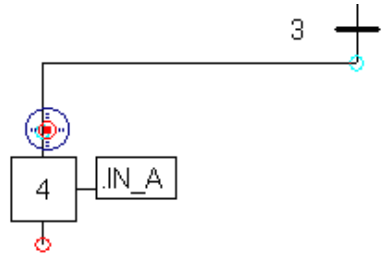


Figure 6-20: Layout of Links - Second connector

4. Click the connector on the final component;

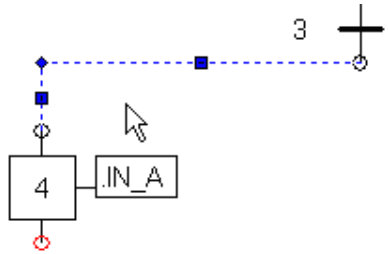


Figure 6-21: Layout of Links - Link created and selected

5. Click an empty place on the document to deselect the link.

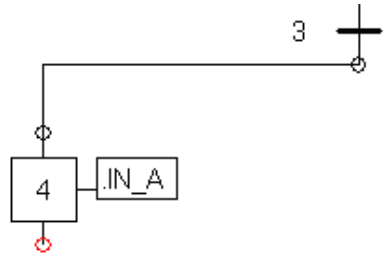


Figure 6-22: Layout of Links - Link created

F The shape of the link is automatically defined by Automation Studio™. To modify the shape of the link, see the *Layout of Links* section in the *User's Guide*.

F A "jump" is a link. If the option "Automatic Jump-to Label on Rising Link" is selected in the SFC options, jumps are automatically created.

6.3.6 Inserting Vergences

Creating a vergence is not more difficult than creating a regular link. Indeed, a regular link is automatically converted into a vergence according to a set of rules.

To create a divergence, you need a SFC with a configuration that is similar to the one below :

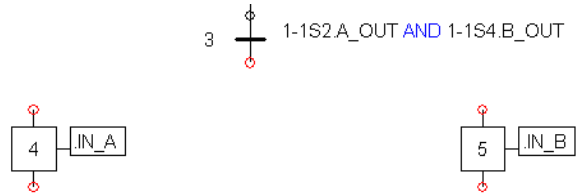


Figure 6-23

The steps are as follows:

1. Link the transition to the first step;

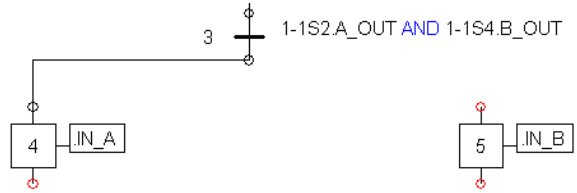


Figure 6-24

2. Link the transition to the second step.

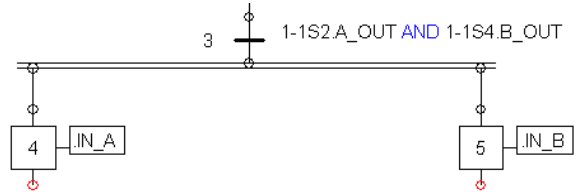


Figure 6-25

The divergence is automatically created. A convergence is created in a similar manner.

You can now complete your SFC as in the figure below.

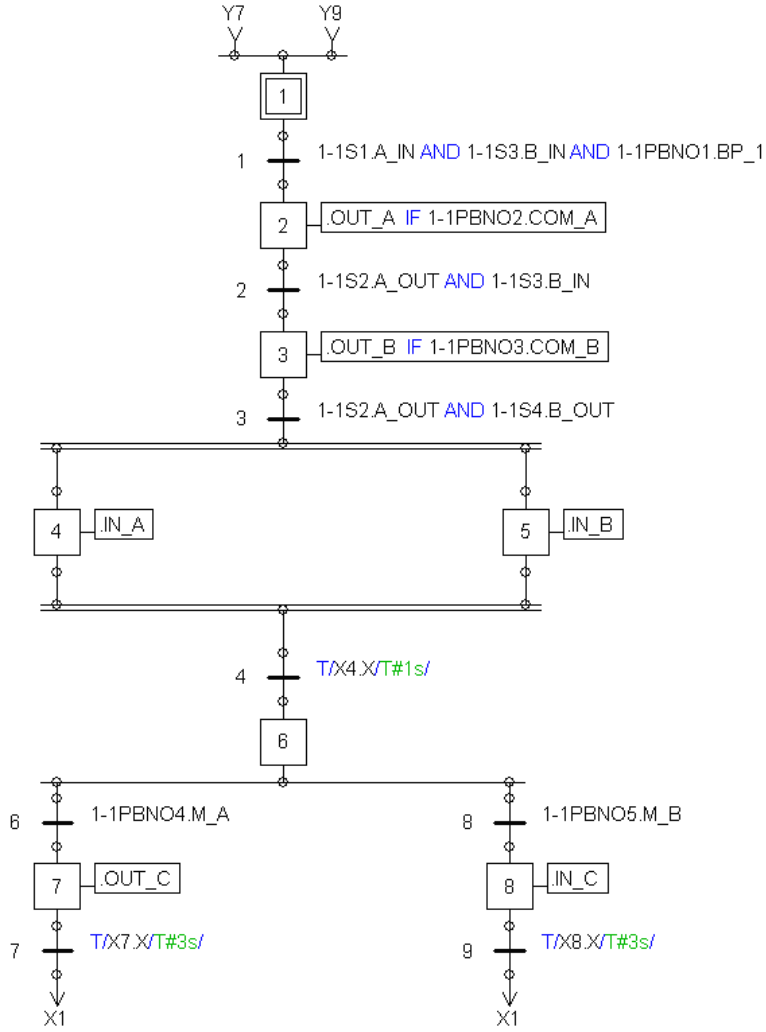


Figure 6-26: SFC with all its links

6.3.7 Inserting Structured Texts

Structured Texts represent autonomous actions that are not associated to a step. These actions are executed when the simulation is active. Structured texts use the same command syntax as do steps and transitions.

To insert a structured text :

1. In the SFC toolbar, click on "Insert Structured Text" (ST);
2. Move the mouse pointer to where you wish to place the new structured text;
3. Left-click. The "Structured Text Properties" dialog box opens.

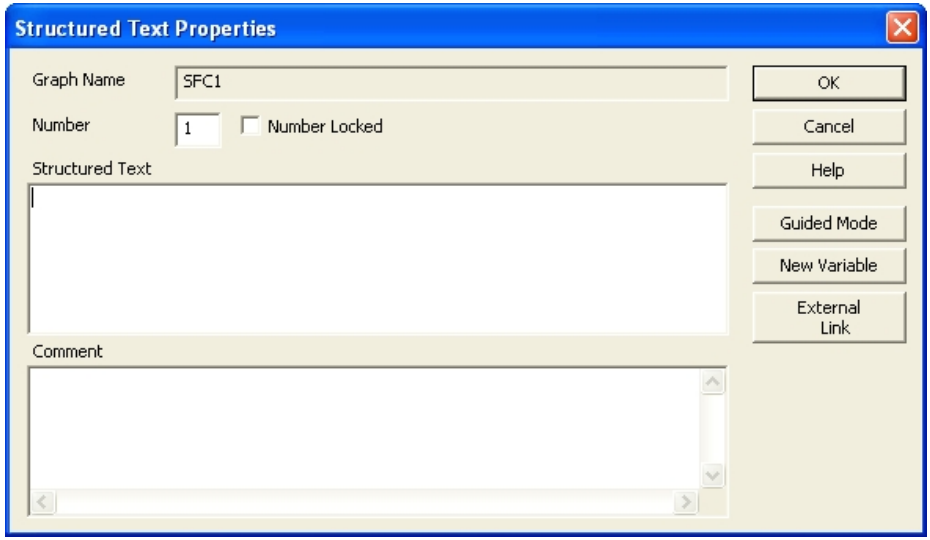


Figure 6-27: "Structured Text Properties" dialog box

4. Click the "Guided Mode" button, and enter the command **.SFC1.RUN := 1**;
5. Enter a comment in the appropriate field. For instance, **Unconditional SFC** run;
6. Click on "OK".

   The Structured Text is displayed.

```
ST1
.SFC1.RUN :=1
Unconditional SFC run
```

Figure 6-28: Example of a Structured Text

6.3.8 Inserting Free Comments

Free comments are not linked to any transition or step.

To insert a comment:

1. In the SFC toolbar, click on "Insert Comment" (C);

2. Move the mouse pointer to where you wish to place the new comment;

   Left-click and the "Comment Properties" dialog box opens.

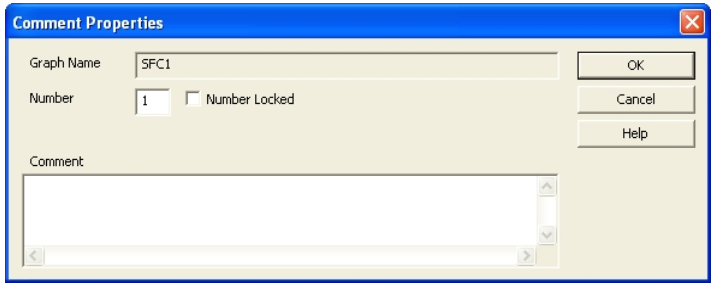


Figure 6-29: "Comment Properties" dialog box

3. For our example, enter **Building a first SFC** in the "Comment" field.

4. Click on "OK".

   The dialog box closes and the comment is displayed.



Figure 6-30: Example of a Comment

6.4 Check for Errors

At any time it is possible to check for errors such as missing links between steps and or transitions, or undeclared variables.

The "Errors…" command is accessible from the "SFC" menu. It allows you to identify components that are not linked, transitions without receptivity, illegal equations, etc. These errors are displayed in a "Messages" window.

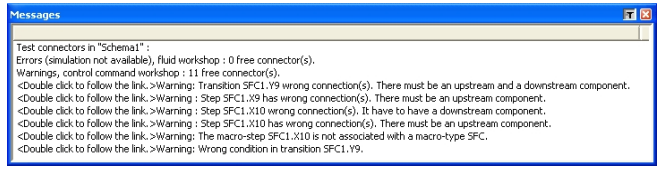


Figure 6-31: Example of errors in the "Messages" window

For more information, refer to the *Automation Studio™ User's Guide*.

7 Simulation

This chapter regroups the particularities associated to the simulation of SFCs. Its purpose is to familiarize the reader with the simulation mode of the workshop. For the various commands related to the simulation of a project or a schematic, see the simulation functions in the *Automation Studio™ User's Guide.*

7.1 State of an SFC During Simulation

During simulation, all states of an SFC having the logical value TRUE are displayed in the head banner of the schematic. These states disappear from the banner when FALSE.

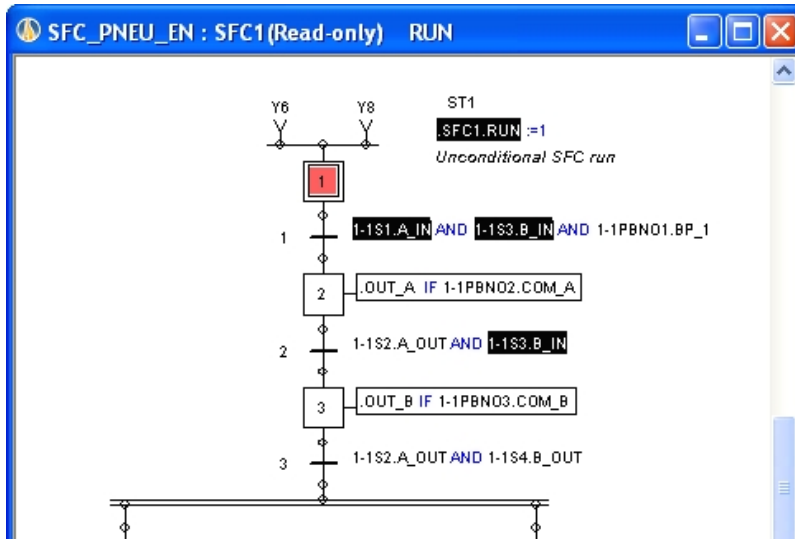
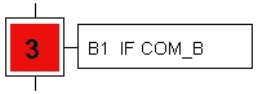
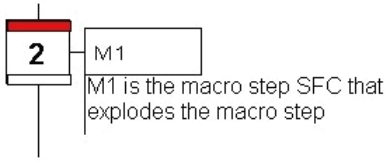
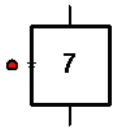
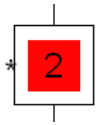


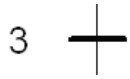
Figure 7-1: An SFC's States during Simulation

7.2 Step and Transition States During Simulation

An activated component is shown with the color that is defined by the "Active Component Color" property. This property can be modified in the "SFC Simulation" branch of the options menu. The default color is red.

The following is a state representation of steps during simulation.

Component	Representation	Description
Active Step		An active step is represented by a fixed red square.
Forced active step		A forced step is displayed as a flashing red square.
Active macro step		An active macro step is displayed as a red rectangle in the upper part of the step symbol.
Active entry step		The entry step of a macro step SFC is activated simultaneously with the macro step of the parent SFC. It consists in fact of an implicit AND divergence.
Active exit step		The exit step of a macro step SFC is deactivated simultaneously with the macro step of the parent SFC. It consists in fact of an implicit AND convergence.
Step with break point		When a step with a break point is activated, the simulation is suspended including all concurrent temporizations.
Step with locked number		When the step number is locked, it is not renumbered if the renumbering function is executed.

Component	Representation	Description
Active transition		A transition with a TRUE validated transition condition.
Active variable		

F Note that a transition is an instantaneous event. Only in the step-by-step simulation mode can it be visualised.

F The variables manager makes it possible to visualise in real time the current value of each variables of an SFC. See the *Variable Manager* in the *Automation Studio™ User's Guide*.

7.3 Fault Management During Simulation

Sequence Duration Surveillance

One sequence duration corresponds to the time elapsed between the deactivation and reactivation of the initial step of an SFC.

Sequence duration surveillances are activated automatically when time limits have been defined in the properties of an SFC.

F Sequence duration surveillance can only be applied to SFCs designed with only one initial step or, in the case of macro steps, with only one entry step. Also, the initial step has to be numbered with the no. 1 step in the SFC.

Minimum Sequence Duration Fault

The elapsed time is cumulated once the SFC has started "RUNNING" and is currently running "RUN". An SFC is considered started "RUNNING" once its initial step is not active. The elapsed time is reset to zero (0) when the initial step is reactivated *X1.X*, or the SFC is reinitialised "INIT". If the SFC returns to its initial step within a minimum time delay, an internal fault is generated and maintained until it is acknowledged.

For this surveillance, we use the following internal variables:

i) SEQ_MIN_FAULT: Minimum sequence duration fault.
ii) SEQ_MIN_ACK: Acknowledgement command for the minimum sequence duration fault.
iii) SEQ_MIN_ACK_MEM: Acknowledgement memory of the minimum sequence duration fault.

For visual acknowledgement purposes, this fault can be assigned to a display light in the Electrical workshop. This assignation can be done inside an SFC or a structured text.

A push button in the Electrical workshop can be assigned to an acknowledgement command of the fault. This assignation can be done inside an SFC or a structured text.

Maximum Sequence Duration Fault

The elapsed time is cumulated once the SFC has started "RUNNING" and is currently running "RUN". AN SFC is considered started "RUNNING" once its initial step is not active. The elapsed time is reset to zero (0) when the initial step is reactivated *X1.X*, or the SFC is reinitialised "INIT". If the SFC returns to its initial step after a maximum time delay, an internal fault is generated and maintained until it is acknowledged.

For this surveillance, we use the following internal variables:

i) SEQ_MAX_FAULT: Maximum sequence duration fault.
ii) SEQ_MAX_ACK: Acknowledgement command for the maximum sequence duration fault.
iii) SEQ_MAX_ACK_MEM: Acknowledgement memorization of the maximum sequence duration fault.

For visual acknowledgement purposes, this fault can be assigned to a display light in the Electrical workshop. This assignation can be done inside an SFC or a structured text.

A push button in the electrical workshop can be assigned to an acknowledgement command of the fault. This assignation can be done inside an SFC or a structured text.

Step Duration Surveillance

One step duration corresponds to the time elapsed between the activation and deactivation of the step.

Step duration surveillances are activated automatically when time limits have been defined in the properties of an SFC.

Minimum Step Duration Fault

For each step under surveillance, the elapsed time is cumulated while the step is active and the SFC is currently running "RUN". The elapsed time is reset to zero (0) when the step is deactivated, or if the fault is acknowledged. If the step under surveillance is deactivated within a minimum time delay, an internal fault is generated and maintained until it is acknowledged.

For this surveillance, we use the following internal variables:

iv) STEP_MIN_FAULT: Minimum step duration fault (only one fault for all steps under surveillance).
v) STEP_MIN_ACK: Acknowledgement command for the minimum step duration fault.
vi) STEP_MIN_ACK_MEM: Acknowledgement memorization of the minimum step duration fault.

For visual acknowledgement purposes, this fault can be assigned to a display light in the Electrical workshop. This assignation can be done inside an SFC or a structured text.

A push button in the electrical workshop can be assigned to an acknowledgement command of the fault. This assignation can be done inside an SFC or a structured text.

Maximum Step Duration Fault

For each step under surveillance, the elapsed time is cumulated while the step is active and the SFC is currently running "RUN". The elapsed time is reset to zero (0) when the step is deactivated, or if the fault is acknowledged. If the step under surveillance is deactivated after a maximum time delay has expired, an internal fault is generated and maintained until it is acknowledged.

For this surveillance, we use the following internal variables:

i) STEP_MAX_FAULT: Maximum step duration fault (only one fault for all steps under surveillance).

ii) STEP_MAX_ACKK: Acknowledgement command for the maximum step duration fault.

iii) STEP_MAX_ACK_MEM: Acknowledgement memorization of the maximum step duration fault.

The following equations are generated automatically and processed implicitly by the software. These variables represent the parameterization for the maximum step duration surveillance:

For visual acknowledgement purposes, this fault can be assigned to a display light in the electrical workshop. This assignation can be done inside an SFC or a structured text.

A push button in the electrical workshop can be assigned to an acknowledgement command of the fault. This assignation can be done inside an SFC or a structured text.

General Fault

An SFC's general fault "GEN_FAULT" regroups all faults declared in the SFC, plus the external security condition "ESC". The software implicitly supports this grouping in the following manner: The external security condition "ESC" can be assigned inside an SFC or structured text.

7.4 Windows Layout

Before switching your first project into simulation mode (*SFC_PNEU.PR5* in the directory *Demos_and_Tutorials* of the software) you have to reorganize all windows in such a way that it is possible to view the operational schematic and the SFC simultaneously.

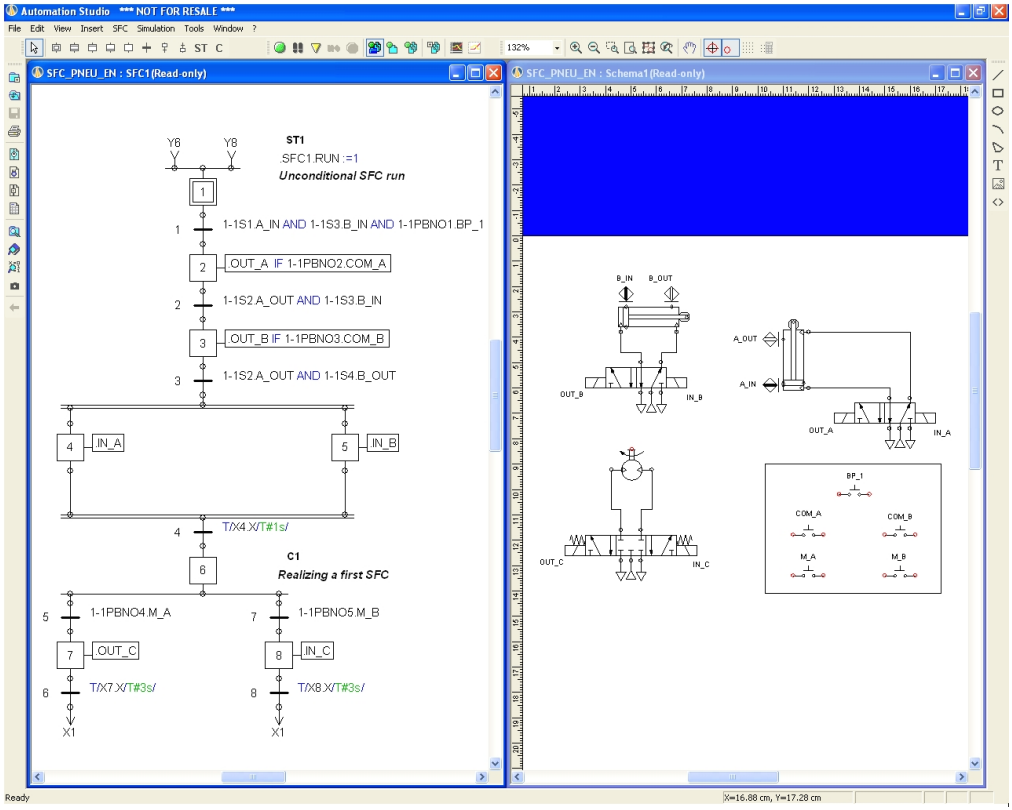


Figure 7-2: Window Layout

7.5 Project Simulation

To manage a project in the simulation mode, refer to the *Automation Studio™ User's Guide.*

7.5.1 *SFC_PNEU_EN.PR5* or *SFC_HYDRO_EN.PR5* Simulation

To simulate the project *SFC_PNEU_EN.PR5* or *SFC_HYDRO_EN.PR5*:

1. Click on the simulation button;

   The SFC is started and the initial step 1 is activated.

2. Click on the push button *PB_1*;

Step 2 is now activated.

3. Click on *COM_A*;

   Cylinder A raises the load.

4. Click on *COM_B*;

   Cylinder B pushes the load on the right.

5. During steps 4 and 5, both cylinders retract to their initial positions;

6. Click on *M_A* or *M_B* to send the load in the left or right containers;

   Step 1 is now reactivated. If push button *PB_1* is clicked again, another load can be processed.

7.6 Possible Causes of Problems

7.6.1 Structure Errors

The SFC editor does not analyse logic errors while an SFC is edited. It limits itself to identifying transitions and steps that have their inputs and/or outputs linked to no other elements in the SFC. In other words, the editor allows creating any SFC imaginable, but with the resulting SFC being not necessarily functional.

Problems can show up when such an SFC is simulated or executed. It is imperative to have the newly edited SFC conform to the norm. When an SFC is in simulation mode, error and/or warning messages are displayed in the "Messages" window, while the simulation is pursued in respect with the fives rules of evolution of SFC. These are as follows:

- The initial situation of an SFC characterises the initial behaviour of the command section with regard to the operative section, the operator interaction and/or the external elements. They correspond to the active steps at the very beginning of all operations: these steps are the initial steps;
- A transition is said to be validated when the steps immediately preceding are all active. The evolution of an SFC can proceed through a transition:
  a) When the transition is validated;
  b) And the associated transition condition has the value TRUE.

- The crossing of a transition simultaneously activates all steps immediately following it,

and deactivates all steps preceding it;

- If multiple transitions are simultaneously validated with the result TRUE, they will all be proceeded through simultaneously;
- If a step is simultaneously activated and deactivated while in operation, the step will be kept active.

7.6.2 Syntax Errors

The editor detects syntax errors related to the steps, the transitions and the structured text. Faulty elements are highlighted in a distinctive colour.

After a simulation is started, error and/or warning messages can appear in the Messages window, but the simulation is carried on. If one of the actions declared in a step is erroneous, all the actions of that step are ignored. If the transition condition in a transition is erroneous or missing, it is always considered as FALSE. If a structured text is erroneous, it is ignored.

A Shortcuts

You can simplify or speed up various operations by using key combinations or key/mouse combinations. This appendix describes all of the key combinations that are available in Automation Studio™.

Appendix A is divided into three sections:

A.1. Keyboard Shortcuts

- Keyboard shortcuts;
- Navigation keys;
- Mouse shortcuts.

Shortcut keys allow you to quickly execute the most frequently used commands. They are generally associated with commands that are common to all the utilities.

File Menu

Shortcut	Command
CTRL+N	New
CTRL+O	Open
CTRL+S	Save
CTRL+P	Print
ALT+F4	Exit

Edit Menu

Shortcut	Command
CTRL+Z	Undo

Shortcut	Command
CTRL+Y	Redo
ALT+ENTER	Properties
CTRL+X	Cut
CTRL+C	Copy
CTRL+V	Paste
CTRL+D	Duplicate
DEL	Delete
CTRL+A	Select all
F2	Allows the editing of text component

View Menu

Shortcut	Command
CTRL+ +	Zoom +
CTRL + -	Zoom -
CTRL+Q	Component snap

Window à "Display" Menu

Shortcut	Command
F6	Opens Toolbar selection box

Shortcut	Command
F7	Opens Message window
F8	Opens Project Explorer
F9	Opens Library Explorer
F10	Opens Plotter
F11	Opens Status bar
F12	Opens Variable Manager

Help Menu

Shortcut	Command
F1	Contextual help

A.2. Navigation Keys

Navigation keys allow you to easily navigate in the screen or through various windows, or to move items.

Navigation Keys Within the Explorers

Shortcut	Movement
#	Go one line up.
$	Go one line down.
PAGE UP or HOME	Go to the top of the tree.
PAGE DOWN or END	Go to the bottom of the tree.

Navigation Keys in the Drawing Editor on a Selected Component

Shortcut	Movement
"	Moves the component one grid step to the right.
!	Moves the component one grid step to the left.
#	Moves the component one grid step up.
#	Moves the component one grid step down.
CTRL + "	Moves the screen view to the right.
CTRL + !	Moves the screen view to the left.
CTRL +#	Moves the screen view up.
CTRL + $	Moves the screen view down.
HOME	Moves the screen view to the upper right corner.
END	Moves the screen view to the lower left corner.
CTRL+HOME	Moves the screen view upper left corner.
CTRL+END	Moves the screen view lower right corner.

A.3. Mouse Shortcuts

Project Explorer

Action	On	Effect
Left click	a diagram	Selects the diagram

Action	On	Effect
Double left click	a diagram	Opens a diagram window

Drawing Editor

Action	On	Effect
Left click	a component in the workspace	Selects the component.
Double left click	a component in the workspace	Opens the component properties window.
CTRL + left click	a selected component	Duplicates the component.
CTRL + left click	a rotation handle	Applies an orthogonal rotation to the component.
CTRL + left click	a resizing handle	Applies a disproportional resizing to the component.
CTRL + left click	a rectangle	Draws a perfect square.
CTRL + left click	a ellipse	Draws a perfect circle.
CTRL + left click	an arc	Draws a perfect arc of circle.
SHIFT+ left click	a non-selected component or in the workspace	Adds the component to the selection.
SHIFT + left click	a selected component	Removes the component from the selection.
SHIFT+left click	a component in order to move it	Moves the component and severs the links.

Action	On	Effect
Right click	a component	Displays the component contextual menu.
Right click	workspace	Displays the diagram contextual menu.
CTRL+scroll wheel	workspace	Zooms in and out the page.
SHIFT+scroll wheel	workspace	Moves the page horizontally.
Scroll wheel	workspace	Moves the page vertically.

B Glossary

Action

An action is an expression that is executed when the step is active. It is an output order from the command part to the operative section. Actions are composed of expressions, assignment operators and command operators.

Every action associated to a step is executed when that step is active.

Active step

An active step (standard or initial) is identified by a red token in the middle of the step symbol.

An active macro step is identified by a red token positioned on top or bottom of the symbol. The token will be positioned on the bottom of the macro step if the out step of the parent SFC is active.

A red token on top of the symbol identifies an active in step.

A red token on the bottom of the symbol identifies an active out step.

Alphanumeric

Characters comprised by the combined set of the 26 alphabetic characters, A to Z, and the 10 numerals, 0 to 9.

AND Convergence

Link between the outputs of at least two steps and the input of a single transition. By definition the AND convergence is a part of the transition.

AND Divergence

Link between the output of a single transition and the inputs of at least two steps. By definition the AND divergence is a part of the step.

Automatism

A device, system, or equipment that allows an apparatus to execute specific operations without the intervention of an operator.

Component

Basic elements used for creating diagrams. Each component represents a behaviour or a function that is functioning during simulation. Components are part of the Library and are provided by the workshops.

Connection

A connection binds two components and/or links them together. Connections are the same colour as the element if they are connected, different if not connected. Two connections from non-compatible technologies cannot be connected together. For example, a pneumatic line cannot be connected to an SFC component and vice-versa.

Connection Number

A number designated for each point where components are connected.

Connector

Element represented by a red circle when it is not linked. It allows the connection of two components. A connector becomes a connection as soon as it is connected to another element.

Diagram

A graphic representation of a circuit or a model by means of elements and components drawn from the Automation Studio workshop libraries. Only SFC components can be placed in an SFC diagram.

Diagram Editor

Tool used to create, modify and manage project files. The project editor includes the document manager that enumerates all the documents in the project.

Display Tools

Designates the design accessories in the Diagram Editor: grid, rulers, connections and connection numbers. You can specify how they are displayed from the View menu.

Edge

Indicates when a Boolean variable passes from one value to the other (0 to 1 or 1 to 0). Edges can be falling or rising.

Edit Mode

Mode of operation in which the components are inserted and changed.

ESC

External Security Conditions that render the running of the SFC useless or dangerous. This condition is used to signal the SFC general fault

Forcing Statement

SFC command which forces a graph to assume a certain situation with active and inactive steps. Forcing statements stay active as long as their condition is TRUE.

The general forcing statement syntax is: F/SFC_Name:(X_i,X_j,…).

GEN_FAULT

Signals an SFC general fault (SFC_name.GEN_FAULT). This signal regroups all graph faults (minimum sequence duration, maximum sequence duration, minimum step duration, maximum step duration) and external security conditions (ESC).

Grid

Horizontal and vertical dotted lines located in the workspace of the editor, used to align the elements of the graph.

Hierarchical Level

Level of an SFC in a forcing hierarchy. The top level is 0.

INIT

Initialization command of an SFC (SFC_Name.INIT) that activates the SFC's initial steps and deactivates all others.

Initial step

Step that is active when the SFC is in an initial situation. The symbol of an initial step has a double border. It is inserted with the use of the Initial Step button in the SFC toolbar.

In Step

The In step is represented by a number inside a square with a rectangle on top. It is inserted with the use of the In Step button of the SFC toolbar. It is used in the SFC that triggers the macro step.

Library

Window containing the basic elements used for designing circuits or simulation models. There are three types of elements: components, links and graphic objects. They are included in the workshops.

Link

A link connects components. During simulation, it is used to transfer information from one component to the other.

Link shape

Path of a link between two points.

Macro Step

The macro step is symbolized by a number in a square with rectangles on both the top and bottom. It is inserted with the use of the Macro Step button in the SFC toolbar. A Macro step is a series of steps/transitions graphically condensed to a single step. It is associated to the SFC that it triggers.

Menu Bar

A horizontal bar located above the application title bar. The menu bar displays the various menus and commands that are available for the active window.

OR Convergence

Link between the outputs of at least two transitions and the input of a single step. By definition the OR convergence is a part of the step.

OR Divergence

Link between the output of a single step and the inputs of at least two transitions. By definition the OR divergence is a part of the transition.

Out Step

The In step is represented by a number inside a square with a rectangle on the bottom. It is inserted with the use of the Out Step button of the SFC toolbar. It is used in the SFC that triggers the macro step.

Project

A group of documents forming a cohesive whole. A Project is managed by the Diagram Editor.

Project Explorer

Content of one of the project editor's windows. The document manager lists all of the documents in the project.

Project Label

Brief description of the project appearing in the project summary.

Properties

Characteristics or parameters of a component. You can change or view the properties by opening the component properties dialog box.

RUN

SFC evolution permission. Without this permission, the SFC is frozen and no steps or transitions are processed.

Sequence Maximum Duration

Time span between the deactivation and reactivation of the initial step of the SFC. When this functionality is used, a fault is generated if the SFC reverts too slowly to its initial step status. This fault must be acknowledged and acquitted.

Sequence Minimum Duration

Time span between the deactivation and reactivation of the initial step of the SFC. When this functionality is used, a fault is generated if the SFC reverts too quickly to its initial step status. This fault must be acknowledged and acquitted.

SFC

Sequential Function Chart (SFC). Graphic representation using basic SFC components used to represent an automatism.

Simulation Cycle

One step in the calculation that determines the state of each component.

Simulation Mode

Mode of operation in Automation Studio during which a project or a diagram is simulated. Unlike the Edit mode, the Simulation mode, when it is active, does not allow any change to the project.

Status Bar

Horizontal bar located at the bottom of all of the utility windows. It contains various types of information such as error messages, zoom factor, mouse pointer coordinates, etc.

Step

SFC element which, when active, executes the actions associated to it. It is represented by a number in a rectangle. It is inserted with the use the Step button in the SFC toolbar.

Step Maximum Duration

Time span between the activation and deactivation of a step. When this functionality is used, a fault is generated if the step remains active for too long. This fault must be acknowledged and acquitted

Step Minimum Duration

Time span between the activation and deactivation of a step. When this functionality is used, a fault is generated if the step is deactivated too quickly, a *Step Minimum Duration* default is generated for the SFC. This fault must be acknowledged and acquitted.

Timers

Timers are processes that produce Boolean results using time such as delay, wait, dead time, etc. as a parameter. The timers used in SFC refer to steps and variables.

Title Bar

Horizontal bar located at the top of a window that contains the title of the window.

Toolbar

A bar located under the menu bar. It contains buttons for the most frequently used commands.

Transition

Must be satisfied (TRUE) to be cleared and to activate the next step.

Transition Condition

A transition condition is a logical condition that determines the evolution of an SFC through a transition. It is a Boolean expression written with the help of input variables, step variables Xn, logical operators, timer operators and edge operators. They can be true or false.

User Interface

Environment made up of windows, dialog boxes, graphic objects, etc., with which the user communicates with the software.

Utility

A general designation for the different types of windows in Automation Studio.

Worksheet

The entire surface available in Automation Studio for generating diagrams.

Workshop

A module that plugs into the Core System. Each workshop contains elements, diagrams and functions relating to its own technological specialty and to the type of diagrams it can generate.

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